

Clinic Report Fall 2022

HOUSING CONNECTICUT

Three projects for affordable housing:
1. Hazel St, 2. Division St, 3. Grand Av

organized by the
Yale Urban Design Workshop

HOUSING CONNECTICUT

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FALL 2022

**1 HOUSING CLINIC
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YALE UNIVERSITY

Executive Summary

Context

Housing Connecticut: Designing Healthy and Sustainable Neighborhoods clinic was an interdisciplinary affordable housing clinic that took place during the Fall 2022 semester at Yale University. It brought together faculty and students from the disciplines of Architecture, Law and Management to collaborate with non-profit developers, creating an unique opportunity to bridge between academia, practice, government, and the local community. The ambition of the clinic was that multi-disciplinary teams of students could combine their expertise and creative problem-solving, to produce new and innovative models for delivering affordable housing.

What the clinic involved

Students were given an intensive “boot camp” style introduction to the world of affordable-housing design and development. Seminars on GIS mapping, data analysis, zoning, building codes, pro formas, financing mechanisms, housing design, and community engagement provided the students with a foundation to work with the developers on new strategic models. Convened by the Yale Urban Design Workshop (YUDW), the Yale School of Architecture’s Community Design Center, in collaboration with the Yale School of Architecture, Yale Law

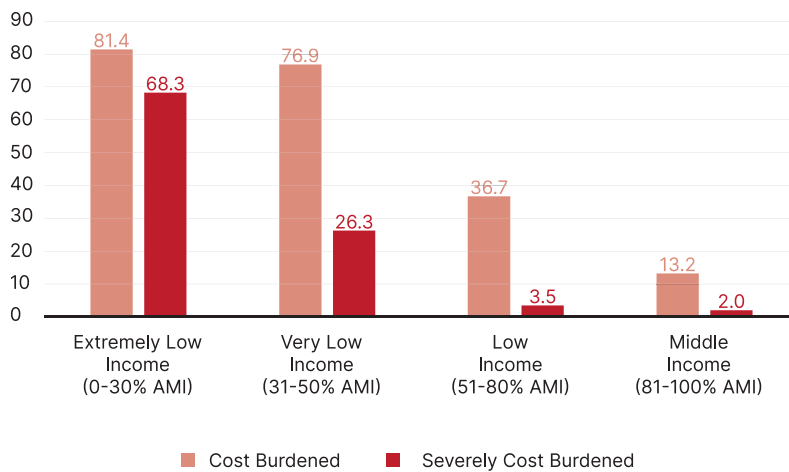
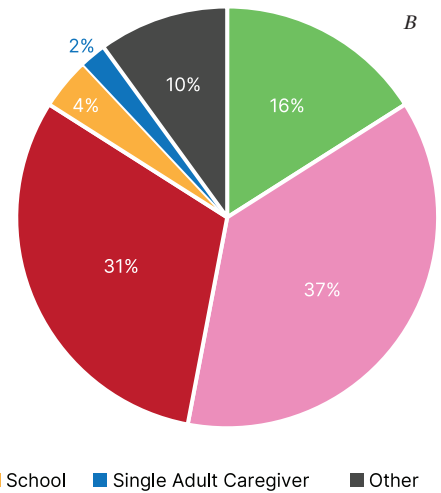
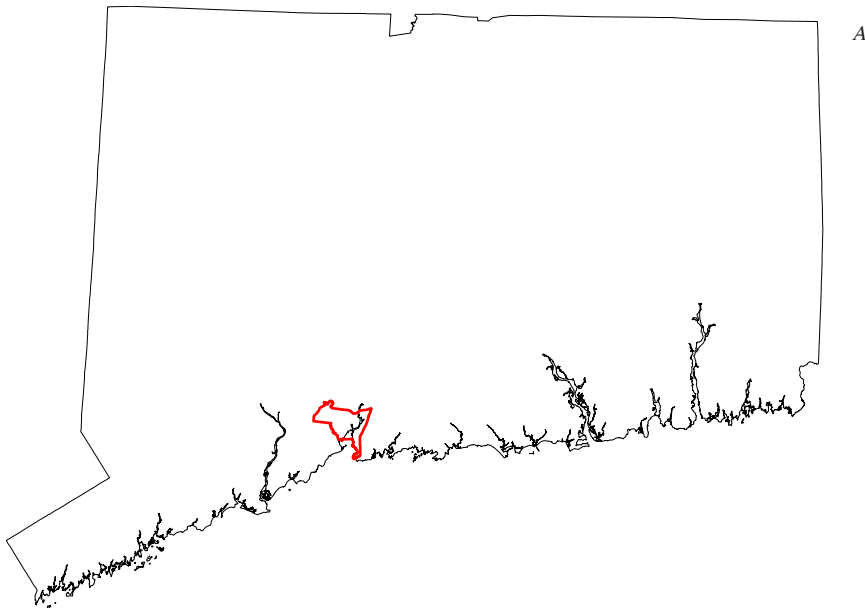
School, and Yale School of Management, along with the Connecticut Department of Housing (DOH), it was the first interdisciplinary clinic of its kind to be taught at the School of Architecture.

Projects that emerged in the clinic

The clinic resulted in three different proposals for affordable housing: two in the New Haven neighborhood of Newhallville and one in the neighborhood of Fairhaven. One project was located on Hazel St in Newhallville and proposed between 8-20 units across two-different schemes, advocating for a block approach to development. The second project was located on Division St in Newhallville and proposed a model based on the potential of convertible ADU housing. A third project for an ambitious mixed-use development was proposed for Grand Ave in Fairhaven.

Future of the clinic

Viable projects emerging from the clinic have the opportunity to be funded directly by DOH and progress towards realization.



Housing Cost Burden by Income Group for Connecticut in 2023 C

Introduction

Fig. A The location of New Haven within the State of Connecticut. Scale: 1:2,000,000

Fig. B Extremely Low Income Renter Households in Connecticut. Source: NLIHC

1 “2023 Connecticut Housing Profile.” *National Low Income Housing Coalition*, 2023.

Fig. C Housing Cost Burden by Income Group for Connecticut. Source: NLIHC

2 “Housing Burden: All Residents Should Have Access to Quality, Affordable Homes.” *National Equity Atlas*, 2020.

Introducing Connecticut

Although Connecticut is one of the wealthiest states in the nation, it is also among the most unequal. This is reflected in a deep housing affordability crisis that is particularly severe for the state’s lowest-income renters. Connecticut has a shortage of more than 89,000 units of affordable housing, with more than 81 per cent of those with extremely low incomes (0-30% AMI) and 76 percent of those with very low incomes (30-50% AMI) being cost-burdened by housing. For extremely low-income renters, there are 61 available affordable homes per 100 renter households. For very low-income renters, there are 37¹.

Connecticut residents of color are particularly harmed by the lack of affordable housing options in the state. 58% of Black and 58% of Latino Connecticut renters are housing burdened, compared to 46% of White Connecticut renters. These numbers are slightly higher than the national rates of housing burden for renters across the United States, where 56% of all Black renters, 54% of all Latino renters and 45% of White renters are rent burdened². These disparities are driven in part by a history of exclusionary housing policies in wealthy suburbs, combined with severe disinvestment within certain municipalities.

Addressing this problem

requires a variety of strategies. Attempts to combat exclusionary land use and zoning policies that hamper the state’s housing production and disincentivize the production of denser, more affordable units are key to both desegregation and the provision of an adequate supply of affordable housing. At the same time, the legacy of car-centered urban renewal and blight removal projects in Connecticut’s central cities means that there is substantial room for infill housing and build-ups in density in these municipalities as well. Both of these strategies are necessary to provide safe, affordable housing for all Connecticut residents, and can also help contribute to more vibrant and connected communities throughout the state.

Introducing New Haven

New Haven is a small, highly urbanized, formerly industrial city in Connecticut with a population of around 135,000 people. While it is home to major employers like Yale University and Yale New Haven Hospital, New Haven exhibits high levels of structural racial segregation and economic inequality due to a combination of historical factors. These factors, including redlining, deindustrialization, urban renewal, and disinvestment have produced a characteristic pattern of population distribution in the state, where the

cities exhibit much higher levels of poverty and concentration of minorities, while the suburban towns tend to be more white and affluent.

Statistically, New Haven as a city is 30% white, compared with 68% in Statewide. Within Connecticut, 10% of people are at or beneath the poverty line, compared with 26% of New Haven's residents. The state has a 67% homeowner-ship rate, while only 28% of New Haven's households own their homes.

The same segregation that occurs across the state is also mirrored within New Haven. Though the city is 30% white, 30% Latino, and 32% Black, that distribution of racial groups is not evenly dispersed in every neighborhood. Newhallville residents are 81% Black and Fair Haven residents are 66% Latino, while East Shore is 67% white.

Though the city overall has a 26% poverty rate, the neighborhood poverty rates range from 3% in East Shore to 43% of residents living in the Hill being impoverished. Meanwhile, 34% of Westville residents are cost-burdened, whereas 63% of Newhallville residents are cost-burdened, relative to the city average of 52%. Across the city, but more acutely in specific neighborhoods, there is a need for affordable housing and robust strategies targeted at the effects of racial and economic segregation.

Housing Connecticut Clinic

The Housing Connecticut: Designing Healthy and Sustainable

Neighborhoods clinic, which took place during the Fall 2022 semester at Yale University, was planned in response to the severe shortage of affordable housing across the state. This new interdisciplinary course paired graduate students from across the disciplines of Architecture, Law and Management with non-profit developers, creating the unique opportunity to bridge between academia, practice and the local community. The intention was to direct the knowledge and enthusiasm being produced within these three disciplines, towards the expertise and networks of existing affordable housing developers and local communities. The ambition of the clinic was that this combination of perspectives would encourage the creation of imaginative models to lessen the challenges associated with housing provision in the current economic climate.

During the semester, students were given an extensive introduction to the world of affordable-housing design and development. Seminars on GIS mapping, data analysis, zoning, building codes, pro formas, financing mechanisms, housing design, and community engagement provided the students with a foundation to work with the developers on new strategic models, identifying sites and preparing design concepts for development. The multi-disciplinary student teams collaborated with three non-profit developers – Neighborworks New Horizons, Neighborhood Housing Services of New Haven, and Beulah Land – to develop detailed proposals across

Fig. D Students from the clinic on a neighborhood walk with Jim Paley of Neighborhood Housing Services, New Haven. Source: Yale Law School

Fig. E Students from the clinic leading a discussion about the projects. Source: Yale Law School

D



E



three different sites in the city. A summary of each of these proposals is included in this report.

Whilst the provision of affordable housing units anchored the design, the projects engaged with issues that cut across community-development, environmental

justice, resilience, social equity, identity, food scarcity, mobility and health. Input from the Connecticut Department of Housing was provided throughout, along with the potential of pre-development support committed, so that the projects may move toward realization.

Neighborhood:

Newhallville

1 Nicholas Dawidoff,
*The Other Side of
Prospect
A Story of Violence,
Injustice, and the
American City* (2022).

Newhallville is a neighborhood within New Haven, bordered by the town of Hamden to the north, by Winchester Avenue to the east, by Munson Street to the south and by Fournier Street on the west.

Although now mostly residential, historically the neighborhood was defined by industry. Winchester Repeating Arms located their factory in the southern part of the neighborhood in 1870 and quickly became a key employer, with their business reaching a peak during World War II by which time they employed around 19,000 people. Much of the housing surrounding the factory was built to house workers and their families, many of whom had migrated from southern states during The Great Migration. From around 1870 to 1920, almost 75 per cent of households in the vicinity of the factory had at least one primary breadwinner employed by Winchester, with other workers engaged in other industrial jobs or commercial activities.

After Winchester Repeating Arms moved most of their production out of New Haven, neighborhood industrial activity dramatically declined. In 2006, the arms factory closed and laid off the small number of remaining workers. The character of the neighborhood gradually changed. As recounted in Nicholas Dawidoff's *The Other Side of Prospect*, "As recently as the 960s, Newhall Street and

Shelton Avenue pulsed with customers walking into the neighborhood's hardware stores, record shops, bakeries, and workingmen's taverns, but by the 2000s its 6500 residents were well accustomed to having no grocery store, no bank, no laundry, no gas station, nowhere to get a money order."¹

At present, Newhallville is a residential, majority-Black neighborhood. It is a predominantly renter neighborhood, though a little over a quarter of residents are homeowners. The median household income is less than \$40,000, which is about three-quarters of the city-wide median, and 60 percent of homeowners in the neighborhood are cost-burdened. Newhallville was significantly impacted by the 2008 housing crisis, it was among the New Haven neighborhoods with the largest proportion of foreclosures and targeted for subprime mortgage loans.

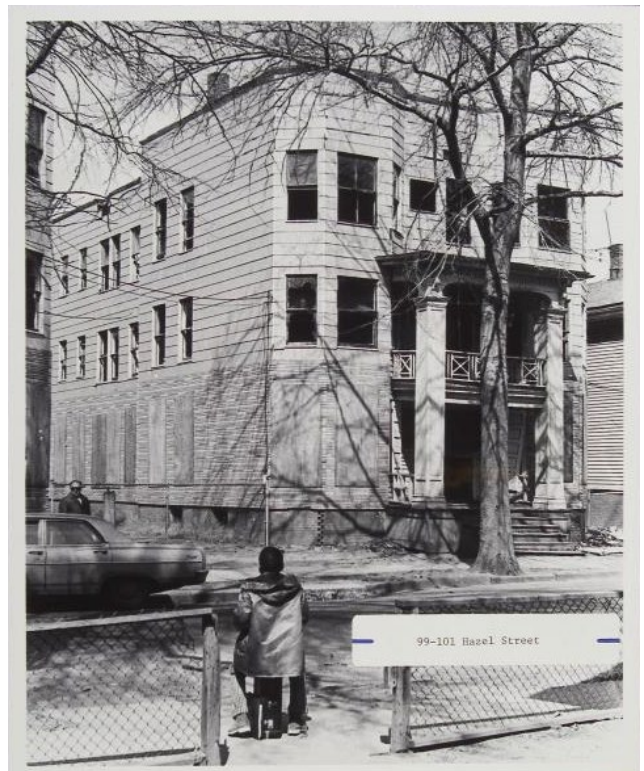
Importantly, a vast majority of the neighborhood's residents believe their neighbors are involved in trying to improve the neighborhood. The neighborhood has both a vibrant past and present community garden tradition, for example on Shelton Ave and Starr St, where the community grows squash, corn, eggplant, green beans and cucumbers.





Fig. A Newhallville in 1959 in a photo taken by the New Haven Re-development Agency. Source: New Haven Museum

Fig. A Newhallville in 1959 in a photo taken by the New Haven Re-development Agency. Source: New Haven Museum



PROJECT:

HAZEL ST

The project proposes a reinvention of traditional residential typologies to provide between 8 and 24 new units of affordable housing for the Newhallville community. In collaboration with Neighborhood Housing Services of New Haven (NHS), local residents and community leaders, the overall strategy of the project addresses multiple scales, from that of the neighborhood, through the support of existing community initiatives, urban improvements and renovations to existing buildings, to that of the individual home, through making use of vacant lots to provide several affordable owner-occupied townhouses.

Team members

School of Management: Chandana Rajanna, Robby Mulcahy

School of Law: Natalie Smith, Nketiah Berko

School of Architecture: Noah Sannes, N'Dos Onochie, Yi Ming Wu

In collaboration with

Neighborhood Housing Services

A



Context

Fig A. Tamika Baines cutting a ribbon on an NHS-renovated West Hazel Street home.
Source: Olivia Gross Photo

Client

Neighborhood Housing Services of New Haven (NHS) is an affordable housing developer in New Haven, led by Executive Director Jim Paley. Their mission is to “strengthen neighborhoods by developing affordable housing and increasing homeownership opportunities; providing homebuyer education and financial coaching; making homes safe, beautiful and energy efficient; and working to improve the perceptions of the communities in which [NHS works] by helping residents take charge of their neighborhoods.”¹ NHS has been working on increasing access to affordable housing since 1979, and has served around 504 families.

Homebuyer education and counseling is an integral component of their work, a service they began offering during the 1990s when house values were declining. These sessions aim to equip residents with tools that would help them face local housing market issues. This led to the creation of the New Haven Home Ownership Center (HOC) in 2001, along with Neighborworks America, which was designed to assist both new homebuyers and existing homeowners. In 1996, they initiated a collaboration with the Yale School of Architecture, constructing a single family home in the Cedar Hill neighborhood. Their present approach

to acquiring properties in order to develop them grew in response to the 2008 housing market crash and the wave of vacant properties that followed. Their current model of neighborhood revitalization in Newhallville focuses on increasing home ownership by developing single or multi-family owner-occupied houses in “clusters” along particular blocks or streets.

For the past 15 years, NHS has acquired distressed properties and empty lots from the City and private sellers at low prices, to develop and then sell as affordable single-family houses. However, this strategy is becoming less sustainable due to changes in the housing market. In light of the economic and political realities, the proposed project aims to offer NHS and the City a collaborative model for development that can offer “missing middle” housing to Newhallville.

Existing Development Plans

The proposal was developed with a full consideration of the planning history of the neighborhood. This involved surveying past and current development visions for both Newhallville and New Haven. The following plans were consulted: Newhallville Renewal and Redevelopment Plan (1968), Newhallville Neighborhood Study (2000), New Haven City Plan Office’s Plan (2003, 2015), The City of New Haven

¹ Neighborhood Housing Services website that says this quote. You can access it here: www.neighborhoodhousing.com. Accessed: 9 December 2022.

Consolidated Plan for Housing & Community Development (2022 Draft for Public Comment), South Central Regional Council of Governments' Municipal and Regional Housing Plans (2022), State Plan of Conservation and Development (2018-2023 Revision), Community Learning Corridor Plan (2018), Dixwell Plaza Plan, NHS Strategic Plan (2019).

The Newhallville community has a strong history of organizing itself to provide assistance to residents and community initiatives that strengthen ties and promote care of the neighborhood. In 2018, community leaders developed plans to turn the plot of land by Hazel and Shelton St in Newhallville, along the Farmington Canal Trail, into a site for education and wellness programming, named the "Learning Corridor". The 2015 City Plan also identified the Farmington Canal Trail as a key vein that connects Newhallville with other neighborhoods in the city. This plan also recommends the "consolidation of vacant and underutilized sites, sliver lots, and/or non-conforming lots to create sizable opportunity for development" in the neighborhood. The proposal for Hazel-Starr St takes on this challenge to think more creatively about how adjacent and smaller-scale plots might be developed concurrently. The State Plan of Conservation and Development (2018-2023) recommends creating "higher density, mixed-income housing stock in areas that are within walking distance to retail, recreation and cultural attractions, jobs and public transit." While

Newhallville is primarily residential, it has the potential to strengthen its links to key recreational areas, such as the Farmington Canal and burgeoning community gardens, by taking on a higher density of permanent residents who can contribute to the use of these sites and community initiatives.

The proposal takes into account the recommendations and suggestions of plans from the state level to the neighborhood level, incorporating them into a spatial strategy that is sensitive to its existing character and scale, whilst ambitious in its vision for the potential of the neighborhood.

Community

The team attended a number of community meetings between September and December 2022 to listen to the residents' concerns and priorities for any development in their neighborhood.

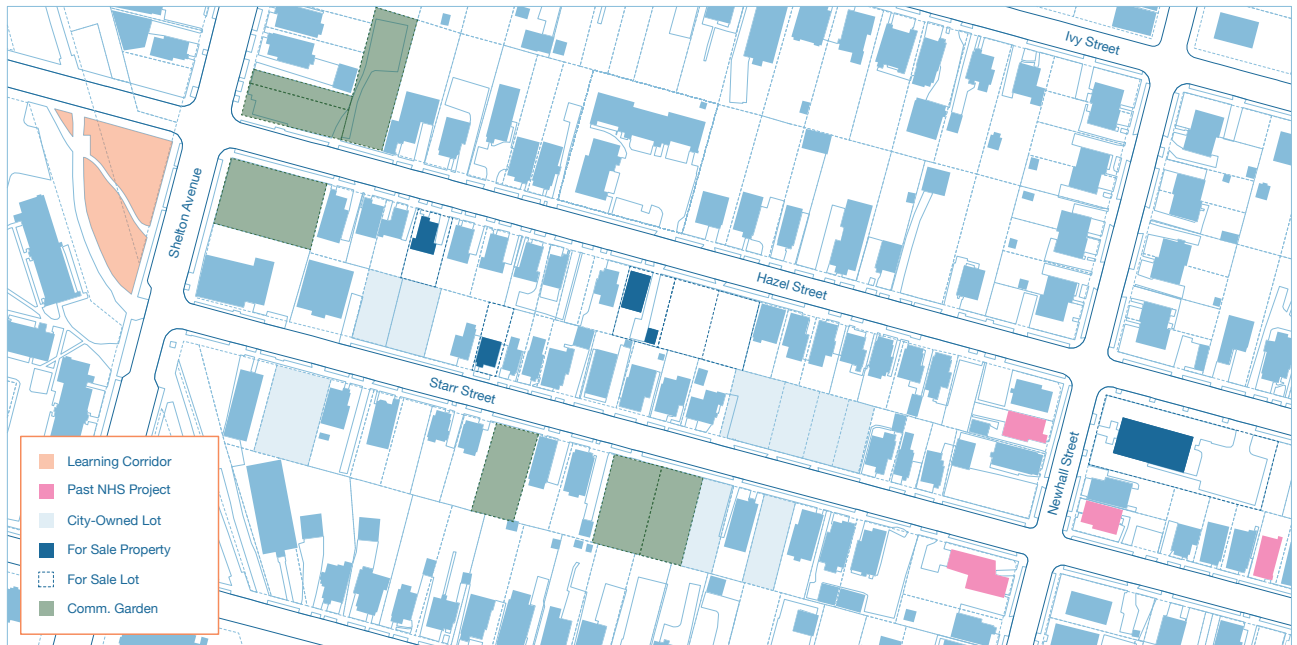
There is a strong tradition of community organizing, particularly in response to improving the neighborhood and making use of its vacant lots. In 1996, The Watson and Bassett Gardening Group (WBGG) formed in response to the deteriorated condition of an open lot at the eponymous street corner. Taken over by weeds, the lot, owned by the City of New Haven, became a dumping ground; despite this, the city ignored resident requests to clean up the site. After a few initial attempts at beautification, five community members organized as WBGG, submitted an application for a Neighborhood Forestry Grant

Fig. B The surrounding block has a varied pattern of ownership, with many of the vacant lots being owned by the City.

Fig. C The site is located close to past NHS projects, the Learning Corridor, and several community gardens.



B



C

from the Urban Resources Initiative, and organized neighborhood meetings to bring in other community members to the effort. Residents pulled weed and picked up garbage, and in the process socialized with each other. The gardening site, located in Block Group 4, was in 1996 surrounded by blocks with

a high rate of homeownership, as well as higher income per capita and median household income.

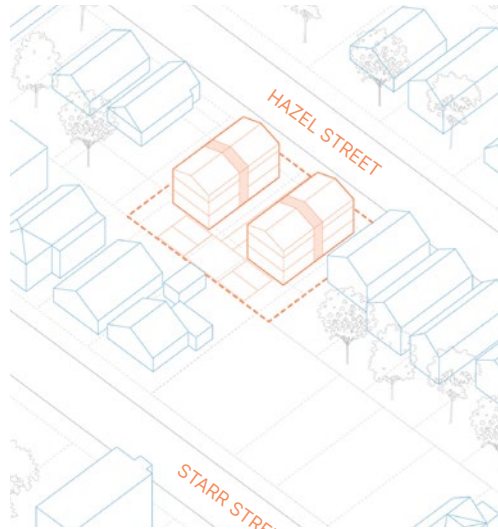
Newhallville has its own Community Management Team (CMT), chaired by Kim Harris. The CMT has monthly meetings in Lincoln-Bassett School.



E

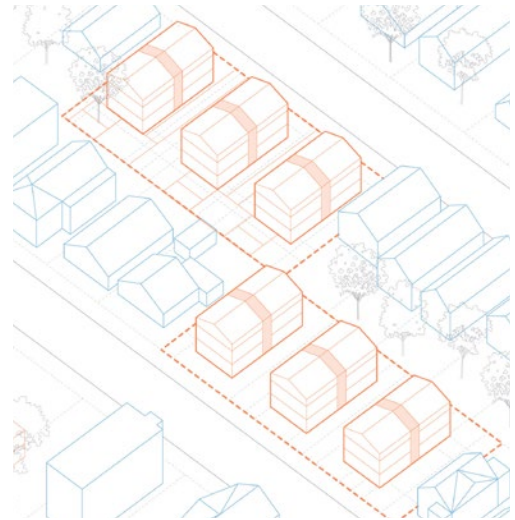
Proposal

Fig. D The proposal has two potential schemes: Scheme A is for two lots on Hazel St and Scheme B incorporates additional lots on Hazel and Starr St.



D

Fig. E Key sites in the neighborhood.



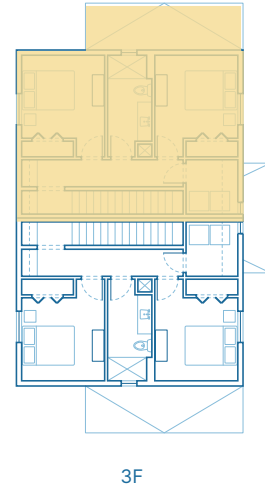
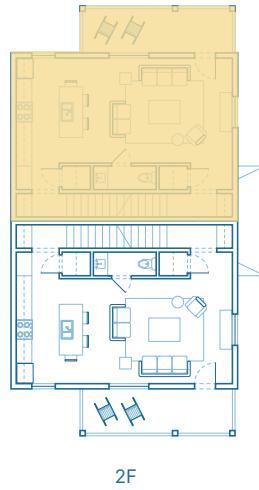
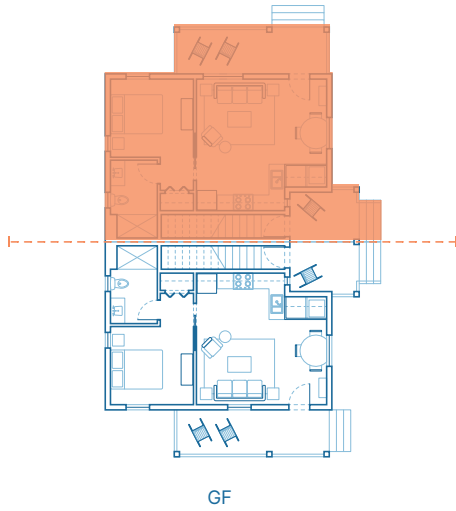
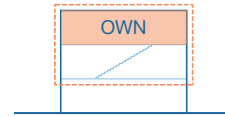
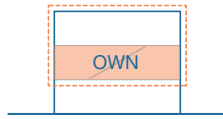
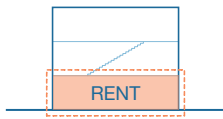
Neighborhood Strategy

At the neighborhood scale, the project proposes improvements to the surrounding area. In order to reinforce existing community initiatives taking place in the neighborhood – such as community gardens and workshops – we propose new public lighting to the Farmington Canal Parkway Trail, covered bus stops near the Learning Corridor on Shelton Ave and a garden shed and new tools for the Shelton-Hazel community garden. We also propose a set of mapping workshops and tools for community members in order for them to understand their own neighborhood, and to plan together to make further changes or to develop new initiatives.

At this scale, we also suggest a block revitalisation strategy

that leverages existing funding sources both at the city and the state level for making improvements to homes that tackle energy efficiency.

A “Zone of Opportunity” exists in Newhallville, centered around the Hazel-Starr block between Shelton Ave and Newhall St. One of the most urgent opportunities is the existence of seven contiguous lots, many of which have sat vacant for decades. Some of these lots have remained vacant due to their dimensions, which, under current zoning conditions, make them hard to develop. However, there exists an opportunity to create an inventive housing model that responds to the condition of the many vacant lots in the neighborhood. The proposal takes advantage of the possibility of developing two adjacent vacant lots at



E

F

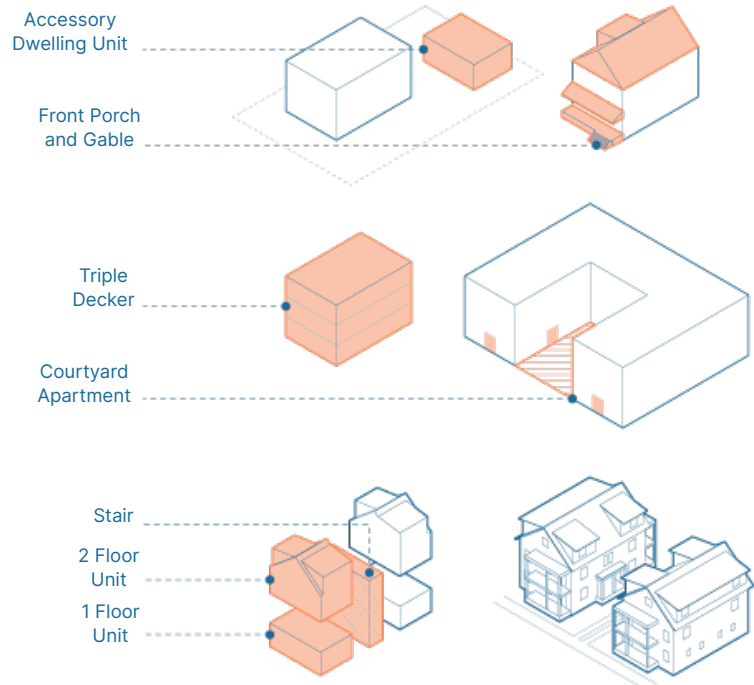


Fig. E Each resident owns one two-floor unit, with a rental unit on the ground floor.

a time, to create several affordable units as well as new and interesting spaces between a building pair.

The proposed site is located close to the Farmington Canal Trail, as well as other community green spaces. Both the latest city plan and community leaders identify this area as a key focus point for the development of the neighborhood and so the chosen site aims to build on work already being done by community activists towards enhancing this area of Newhallville, as well as NHS's desire to continue development along or close to Shelton Ave.

Site Strategy

The proposal presents two options for development. Concept A, Hazel, proposes the development of two vacant lots located on Hazel St, currently on the market for private sale. Concept B, Hazel and Starr, proposes the development of these two lots in tandem with the development of four city-owned vacant lots on Starr St.

The architectural strategy was informed by the scale and quality of existing housing typologies which define the fabric of the neighborhood. These include elements such as front porches, triple deckers and the emergence of Accessory Dwelling Units. The final proposal consists of building pairs that combine two two-family units that draw upon and reinvent these spatial features.

Each two-family unit resembles a typical three-storey New Haven house, complete with a pitched

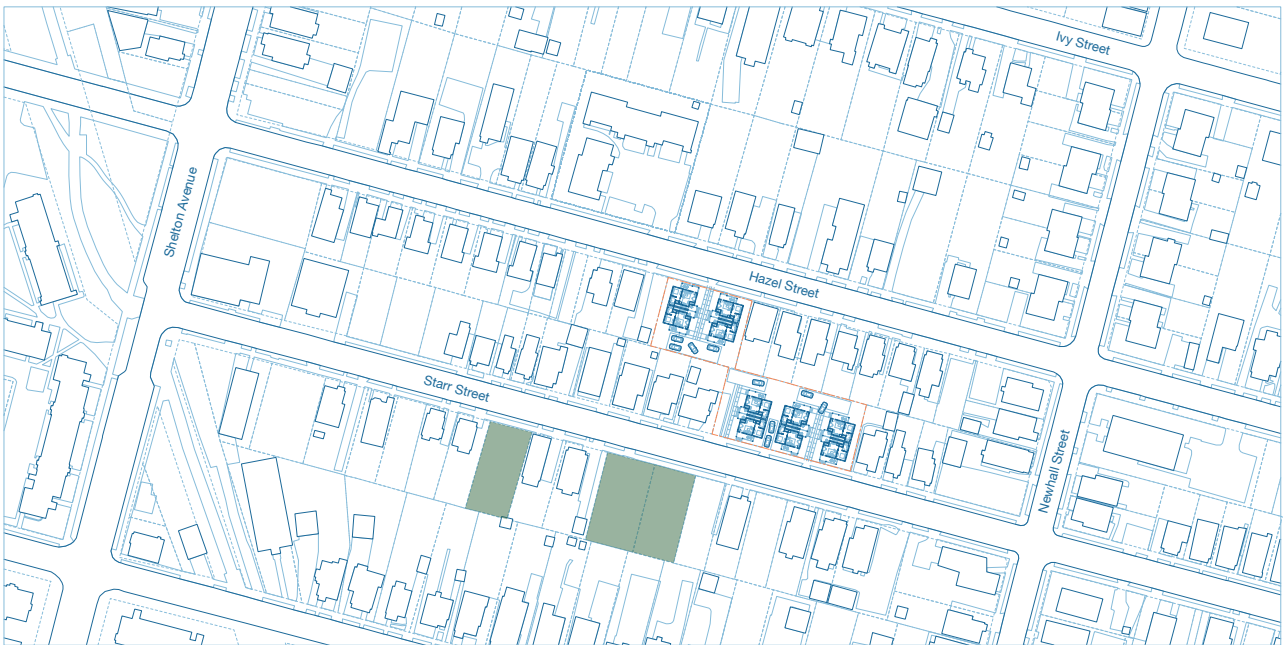
roof, to fit the character of the area. The unit includes a ground-floor one-bedroom unit that can be rented out by each owner-occupant living in the two-story, two-bedroom unit above it. The two-family units are divided front to back, instead of side-to-side, as in a traditional rowhouse.

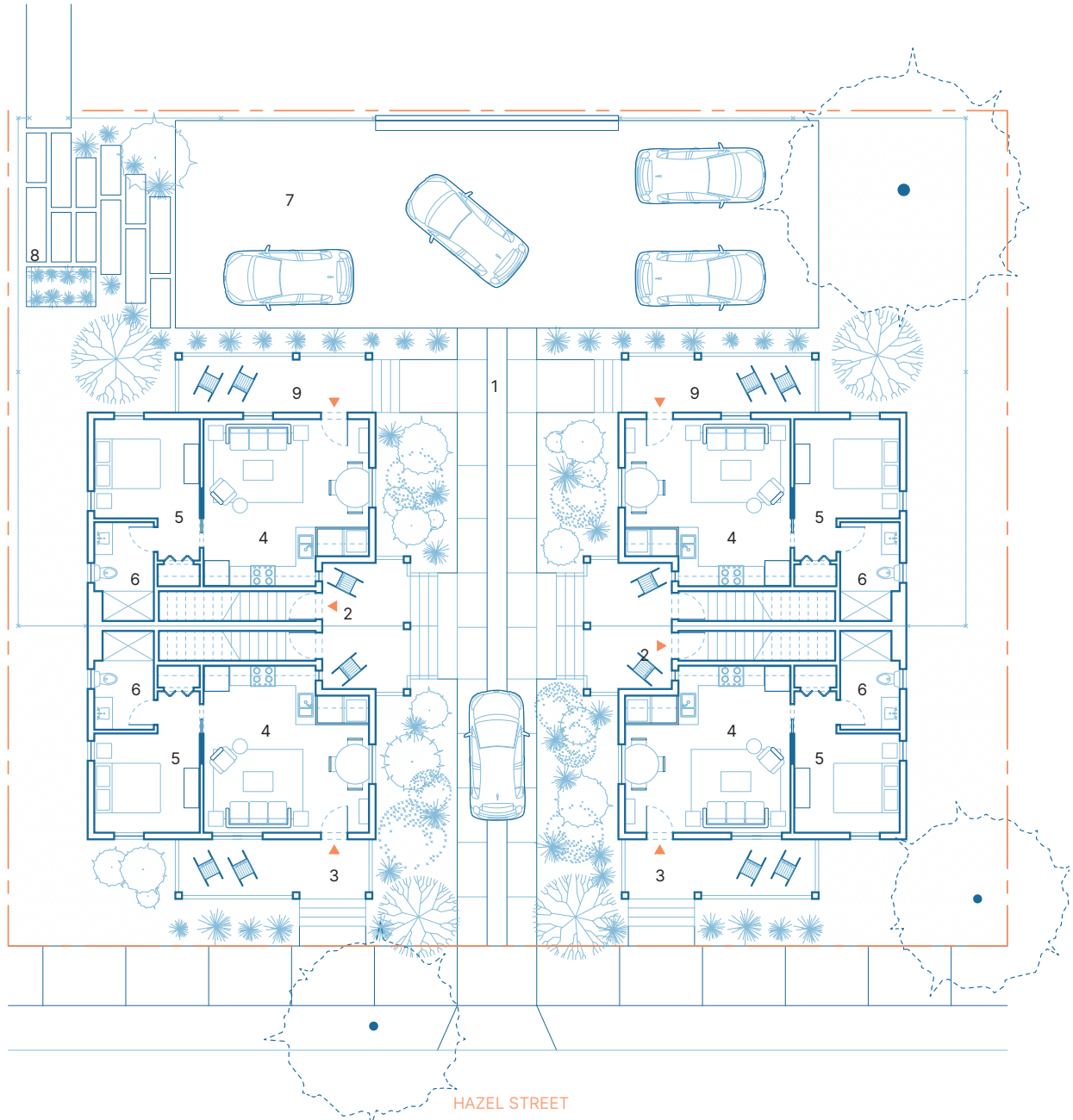
The ground-floor one-bedroom unit is accessed directly from Hazel St, with the other units accessed off a pleasantly-landscaped alley between the building pair, carefully designed to accommodate pedestrians and cars. This configuration ensures the spaces between the buildings have good visibility in all directions, something requested by the Newhallville residents in various meetings. The design of the porch-covered entrances also means the center of the alley operates like an open courtyard, providing an intimate space to pause, for neighbors to gather and to interact.

The townhouse is designed to utilize Passivhaus design strategies, such as solar panels, operable windows for ventilation, and porches for sun-shading, with the goal to create energy efficient buildings that also helps to reduce the burden of energy bills on the resident.

Fig. F The new typology borrows key characteristics from other residential buildings in the neighborhood.

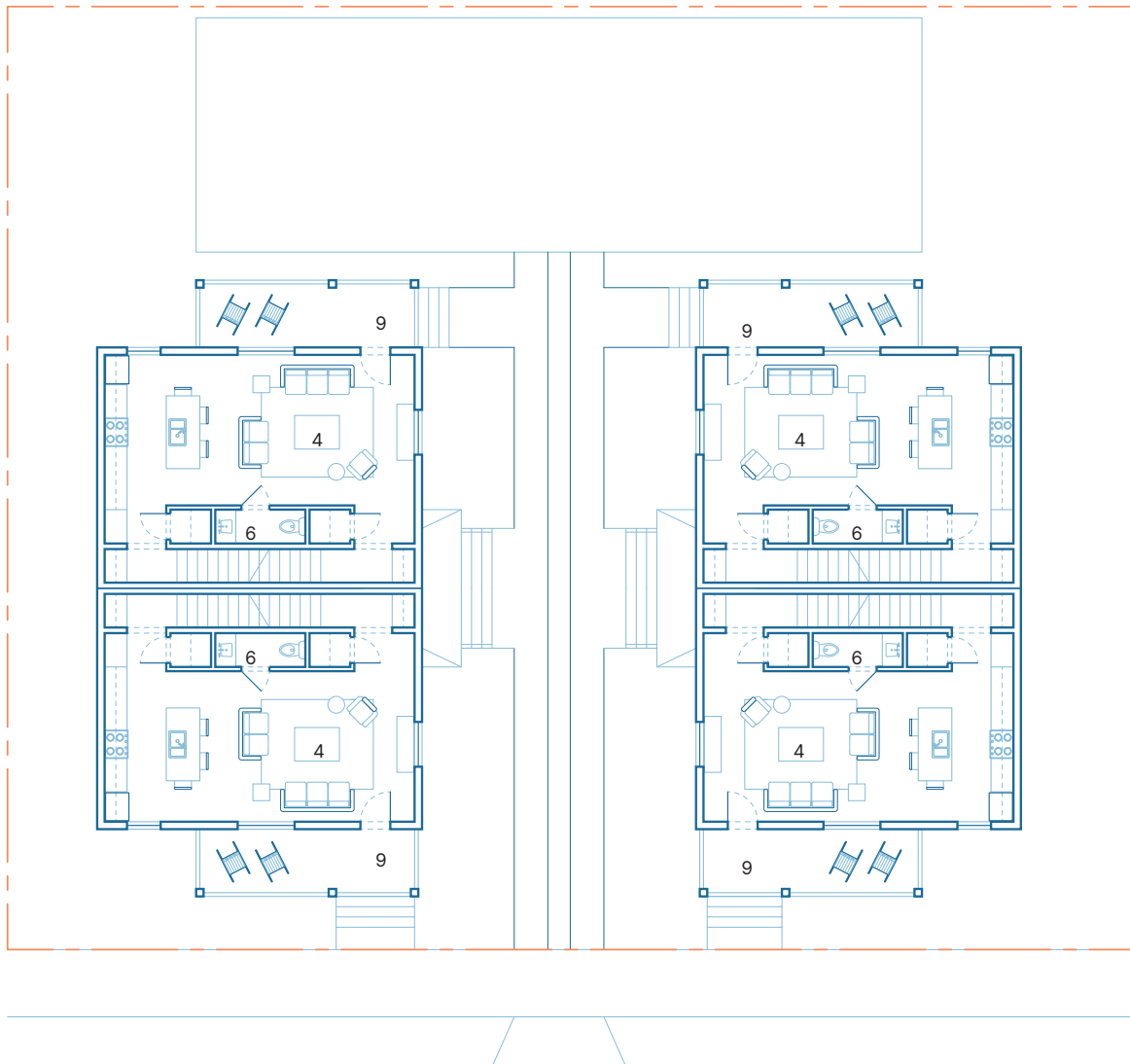
SITE PLAN





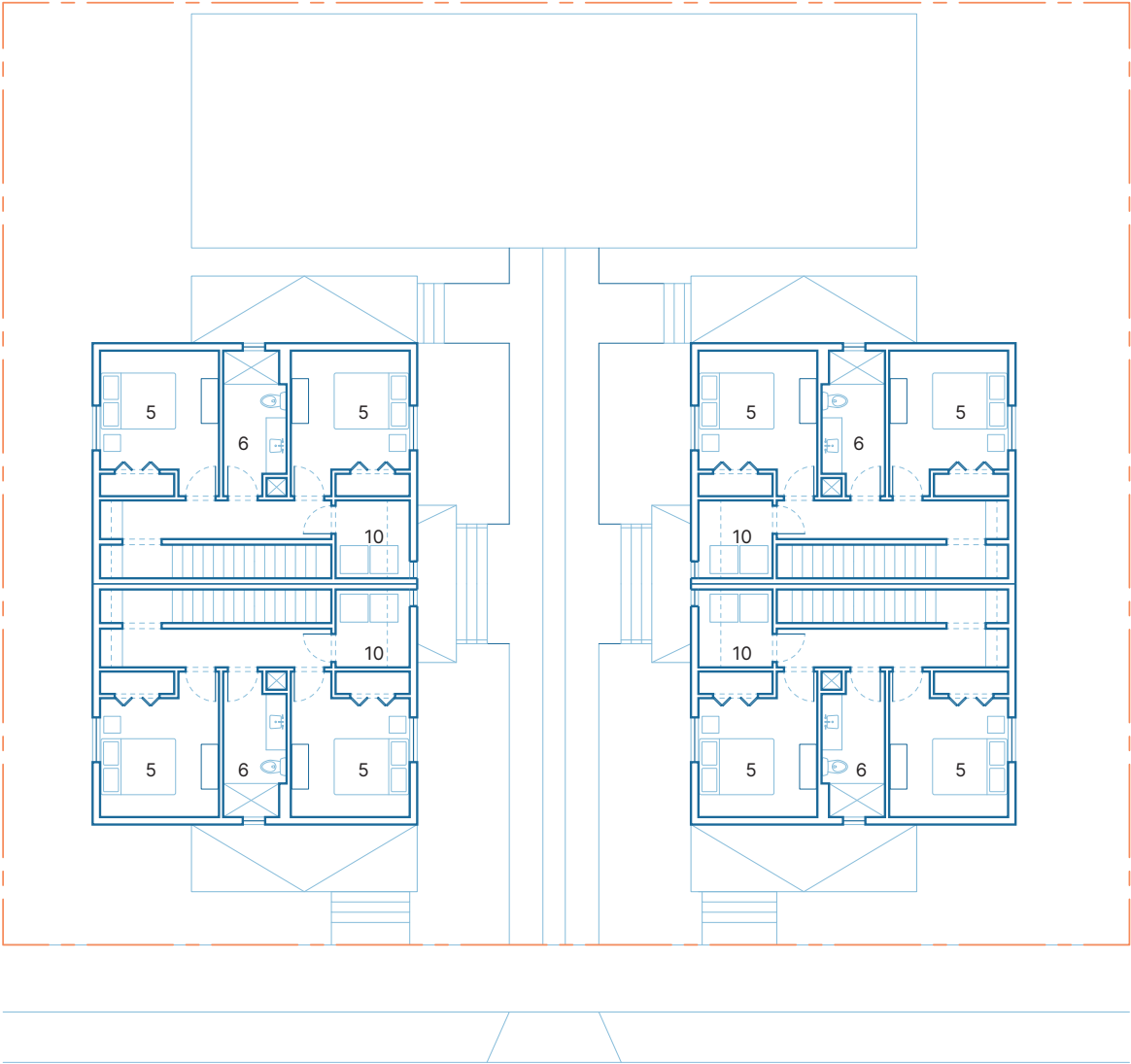
- 1 Pedestrian Alley
- 2 Owner Entry
- 3 Tenant Entry
- 4 Living/Dining
- 5 Bedroom
- 6 Bathroom/WC
- 7 Parking
- 8 Community Garden
- 9 Balcony





- 1 Pedestrian Alley
- 2 Owner Entry
- 3 Tenant Entry
- 4 Living/Dining
- 5 Bedroom
- 6 Bathroom/WC
- 7 Parking
- 8 Community Garden
- 9 Balcony
- 10 Washer/Dryer







A new pathway is created through the block, taking advantage of an overlap between two parcels of land on Hazel and Starr.



SCHEME A

UNIT MIX

Affordable Housing Units	Studio		1 Bedroom		2 Bedroom		3 Bedroom		Total
	Units	Monthly Mortgage* Payment	Units	Monthly Mortgage Payment	Units	Monthly Mortgage Payment	Units	Monthly Mortgage Payment	
30% AMI (Deeply Affordable)	-	-	-	-	-	-	-	-	0
50% AMI	-	-	-	-	4	\$1390	-	-	4
60% AMI	-	-	-	-	-	-	-	-	0
Total	0		0				0		4

*Mortgage assumes 5% down payment, 7% interest rate, 30-year mortgage term

Financial Strategy

A financial plan is provided for Scheme A. The affordable units will be constructed for sale, to be purchased by potential owners with income less than 80% of AMI for Newhallville, at a cost of approximately \$240,000. The average sale price for homes in Newhallville is \$290,000. Homes sold by the Liveable Cities Initiative (LCI) sell at around \$215,000. NHS has previously worked with a 100% equity structure and therefore for this project there will be no debt component. Four sources of capital for the project have been taken into account: state tax credits, department of housing funding, sales proceeds and federal HOME program grants. The uses for these funds are spread between land costs, hard costs, soft costs and developer fees.

The townhouse model of ownership has been proposed. Subdividing the lots into smaller, non-conforming lots along the party walls will allow owners to circumvent common interest ownership for shared elements such as the driveway.

Our model anticipates that resident-owners will rent out the first floor of their unit. In this scenario, NHS will provide training to ensure the resident-owners are aware of their responsibilities for the property and future tenants.

For the units to be affordable, the sale price unit should be accessible to owners with an annual income less than a region's 80% median income. In targeting low-in-

come buyers, an affordable purchase price can be arrived at using one (or both) of two methods:

Method 1: Starting with area median income (AMI) to determine an affordable mortgage amount; In this approach, a 4-person household earning 80% of AMI can afford a \$267,700 mortgage assuming 27% of the household income (\$79,900) is spent on mortgage payment for a 30 year loan with 7.0% annual interest.

Method 2: Starting with a target mortgage amount to determine affordability to low income purchasers: In this approach, if we consider a purchase price of \$200,000, then assuming a 30 year mortgage with 7% annual price and 27% as the % of income spent on mortgage, then we arrive at a figure of \$59,694 for the annual household income.

The average sales price for homes in Newhallville is \$290,000 as per numbers published on Redfin. Livable cities initiative (LCI)'s affordable homes sell at a price of \$215,000.

Therefore, for the potential owner, the expected mortgage burden annually is \$16,117. As the units are two family houses, the owner can expect an annual rental income of \$8,400 (at \$700 monthly rent) for the second unit and therefore net annual expenditure would reduce to \$7,717 (\$16,117 - \$8400).

SOURCES AND USES

Sources		Uses	
City HOME Funds [a]	\$400,000	Land Acquisition [e]	\$100,000
HTCC Subsidy [b]	\$100,000	Closing Costs	\$2,000
Sale Proceeds [c]	\$880,000	Hard Costs [f]	\$1,628,000
Dept. of Housing Grants [d]	\$587,744	Soft Costs [g]	\$68,131
		Contingency	\$84,807
		Developer Fee	\$84,807
Total	\$1,967,744	Total	\$1,967,745

[a] Assume \$100,000 grant per unit

[b] Assume \$25,000 grant per unit

[c] Assume \$220,000 sale price per unit based on historic NHS sales at around \$240,000 per unit and Livable Cities Initiative's affordable units at Thompson Street sold for \$215,000 per unit

[d] Calculated as gap between development budget and sum of other sources

[e] Land Acquisition Price: Based on listed price

[f] Hard Costs: Assume \$220 per square foot

[g] Soft Costs, Contingency, Fees, Miscellaneous Expenses: Assume 10% of total budget

CASH FLOW SUMMARY

STABILIZED YEAR

Developer Revenue		2024
	<i>Sale Price</i>	
2BD House Sale	\$220,000	\$880,000
Gross Rent Revenue		\$880,000
Less Brokerage Fees	2%	\$862,400

Rent Escalation	0.02
Expense Escalation	0.03

<i>Mortgage Assumptions</i>	
w/ PMI	Y
Down Pmt %	5%
Term (years)	30
Interest Rate	7%
Sale Price	\$220,000

	<i>Monthly Payment</i>	
Annual Mortgage	\$1390.482	\$16,685.787
PMI	1.21%	\$201.898
Property Taxes	3.975%	\$6996
Home Insurance	\$110	\$1359.6
Total Annual Payment		\$25,243.285
Water (both units)	\$70	\$865.2
Utilities (owner unit)	\$75	\$927
Capital Reserves	\$66.667	\$824
Landscaping/Snow Removal	\$30	\$370.8
Annual Rental	\$1000	\$12,240
Vacancy	0.05	\$11,628
Total Annual Payment		\$16,602.285
		\$1383.524





PROJECT:

DIVISION ST

The project is a proposal for 27 units of housing on a difficult site in a transition neighborhood in New Haven. The apartment building makes use of new laws surrounding ADU's to provide an affordable and flexible model of home ownership, with the intention of attracting and supporting residents to remain in the neighborhood long-term. Bridging scales, the design of the housing draws from the surrounding typologies and architectural language of the neighborhood.

Team members

School of Management: Khayla Smith, Brandon Jones

School of Law: Dominique James, Raymond Fang

School of Architecture: Zishi Li, Calvin Liang, Mariel Lindsey

In collaboration with Beulah Land



A

Context

Methodology

To engage and research the community, six key strategies were utilized to analyze and assess community needs.

In September, the team joined Beulah for a walking tour of the Dixwell neighborhood. Beulah showed the team several of the single-family homes they've developed and their recent expansion into affordable multi-family rentals in Dixwell. eam mapped out New Haven non-owner-occupied housing in the Newhallville area, focusing on the blocks surrounding 51 Division Street. We found a high concentration of non-owner-occupied housing, with many of the properties owned by local slumlords or individual Landlords. TheThen, the team attended the October meeting of the Newhallville Community Management team. The CMT was receptive to exploring nonprofit affordable housing development in partnership with Yale.

In November, the team met with Kimberly Edwards, the Alder for Ward 19 in New Haven. Ward 19 contains a portion of Newhallville, including 51 Division Street, as well as Prospect Hill and East Rock. Alder Edwards expressed a strong preference for homeownership of affordable units over affordable rental housing.

In November, the team researched and mapped incoming

housing developments in Newhallville to identify gaps in the types and prices of new housing being provided. In November, the team attended a Newhallville community meeting organized by Science Park Development Center, Twining Properties, and LXMD. The developers gave updates on their housing and real estate development at the Winchester Center. Our team listened to resident concerns and feedback over this development.

Based on this community engagement process, the team identified three critical needs in the Newhallville community: (1) provide a diversity of housing options; (2) stabilize the neighborhood; and (3) address the racial wealth gap. As a result of this analysis, the project will meet Newhallville's need for housing options through sustainable, affordable homeownership to address New Haven's racial wealth gap.

The first project goal is to build affordable family housing to meet Newhallville's housing needs where the market is not. BLDC and attendees at the Winchester Center community meeting expressed concern that the incoming housing developments were not meeting the affordable family housing needs of Newhallville residents. The analysis of Census data indicated that Newhallville has, on average, larger families of lower income living in poverty compared to New

Fig. A The leafy site offers the potential to accomodate different scales of building.

Haven. Incoming developments are not meeting this need. At least 843 luxury units across three developments have been or will be built in the southeast corner of Newhallville in the next few years. Nearly 90% of these units are market rate, and the majority of the units are studios and one-bedroom units. Only 89 affordable units will be built in the same period of time. Clearly, these units fail to meet the need for affordable

Client

Beulah Land Development Corporation (BLDC) was established in 1994 emerging out of the Beulah Heights 1st Pentecostal Church community. BLDC is a faith-based non-profit, focused on developing affordable housing for low to middle income tenants. BLDC views its work as an extension of the Civil Rights Movement. Their mission statement explains: "We aim to rebuild in New Haven's most underserved communities, developing revitalized communities of choice for residents, opening doors to homeownership, creating job opportunities, and fostering economic development."

Thus far, BLDC has been able to effectively execute their mission and find success within Dixwell. As a result, the team focus was largely to assist BLDC in extending beyond their present Dixwell boundaries and expand their reach and vision of home ownership into Newhallville.

Proposal



Fig. B There are several different types of unit in the development.

Neighborhood Strategy

This scheme has the potential to expand to other properties/territories. Potential sites of development include 1) 310/312 Dixwell Ave, owned by the City of New Haven, 2) 777 Orchard St, owned by the City of New Haven and 3) 771 Orchard St., owned by Beulah Land.

These prospective sites of development all share similar traits to that of 51 Division St., and allow the scaling of the project's design method and strategy to other sites in Dixwell and Newhallville to provide opportunities of affordable home ownership for the community.

Architectural Strategy

The site is the lot of 51 Division Street, is currently owned by the Housing Authority of New Haven. The existing parcel is currently zoned as a PDU-21, which is a variance upon a typical RM-2 residential zoning. The PDU-21 variance allows for a maximum of 36 units of

housing on the site given 1:1 parking per unit. The site sits strategically on the border of the Science Park development across Division St. to the south, and the greater Newhallville community to the north. To gain site control within the next calendar year, a co-developer agreement or ground lease agreement with HANH and Beulah would be required.

The existing parcel is currently zoned as a PDU-21, which is a variance upon a typical RM-2 residential zoning. The PDU-21 variance allows for a maximum of 36 units of housing on the site given 1:1 parking per unit. The massing strategy takes into account the depth of this particular lot, (which is approximately 500 feet deep). The back of the site lends itself to kindly to placing on-site parking, which is within the given requirements of the PDU-21 zoning., which we are currently not aiming to apply for variances on. An extended drive runs from Division St to the back of the site in order to provide access to the parking and to create

SITE PLAN



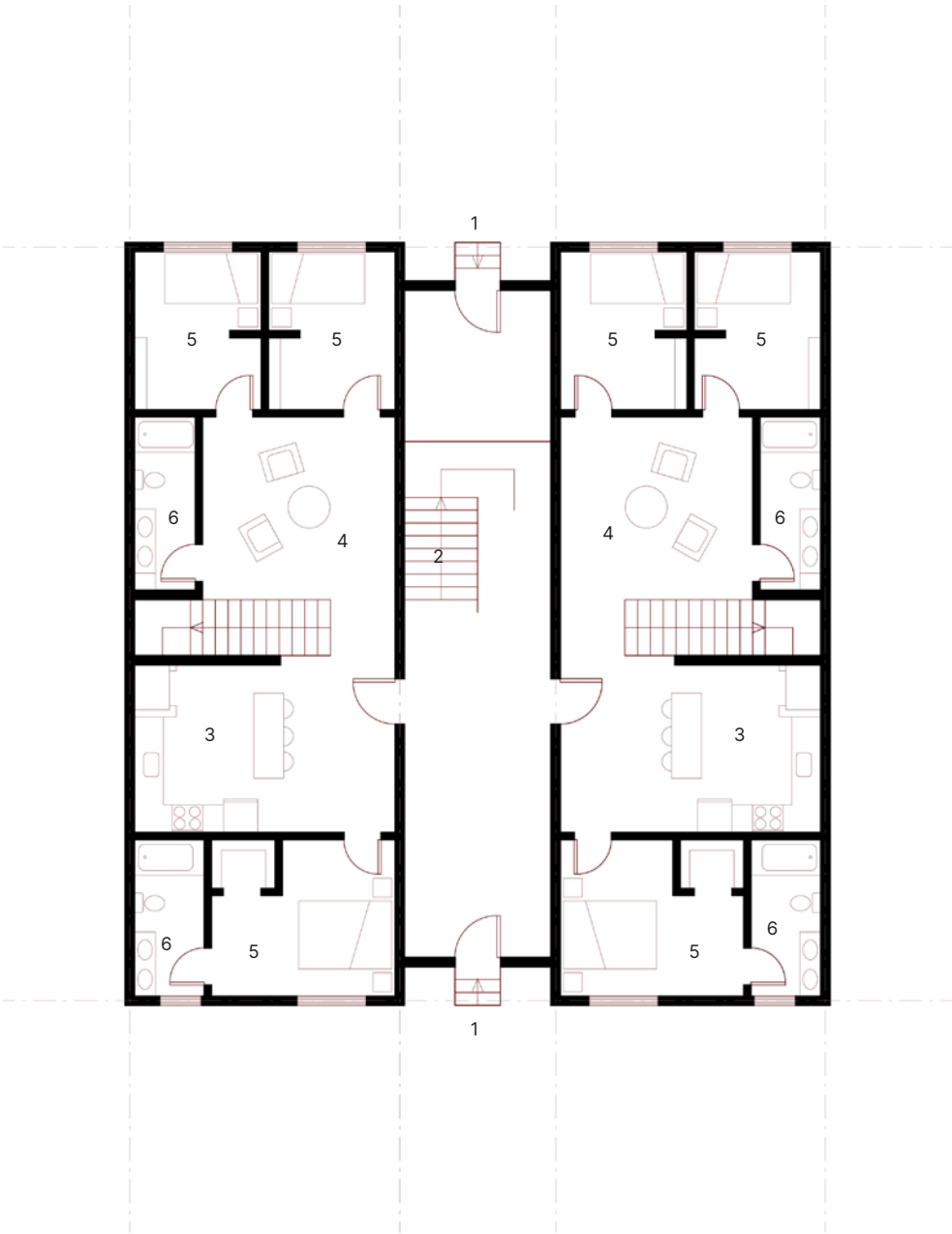
a mews-like feel. In order to access this back of the site we propose an extended drive which runs from Division St. to the rear of the site. This provides the opportunity to incorporate into the site smaller units to the West, providing a varied range of unit-types, and which To buffers theis site drive from the ADU's in the backyards of the homes on Sheffield St. to the west, the project incorporates a series of garage and garage rental units.

The primary massing which is approximately about 300 feet deep and 60 feet wide, and incorporates allows a variety of housing options, including ., but ultimately the project pursues family apartments, with a mix of studio and 3-bed units in order to be responsive to the needs of community members desires.

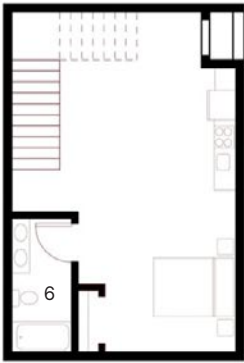
The site is bordered to the west by a vernacular house with sloping roof and wood siding on the west, and a three-storey brick an apartment complex to with brick row houses on the east. The many landscape and elm trees along Division Street also give the site a unique character. The project is We hope that the designed to be sensitive to this context, unveiling itself humbly unveils itself from behind the canopy of the existing elm trees. for pedestrians and drivers alike. The townhomes and ADUs may have facades that connect the context on both sides along Division Street—for example, the ADUs may bear matte metal cladding facades while the townhomes have a brick facade facing Division Street. A There is a shared lobby is added

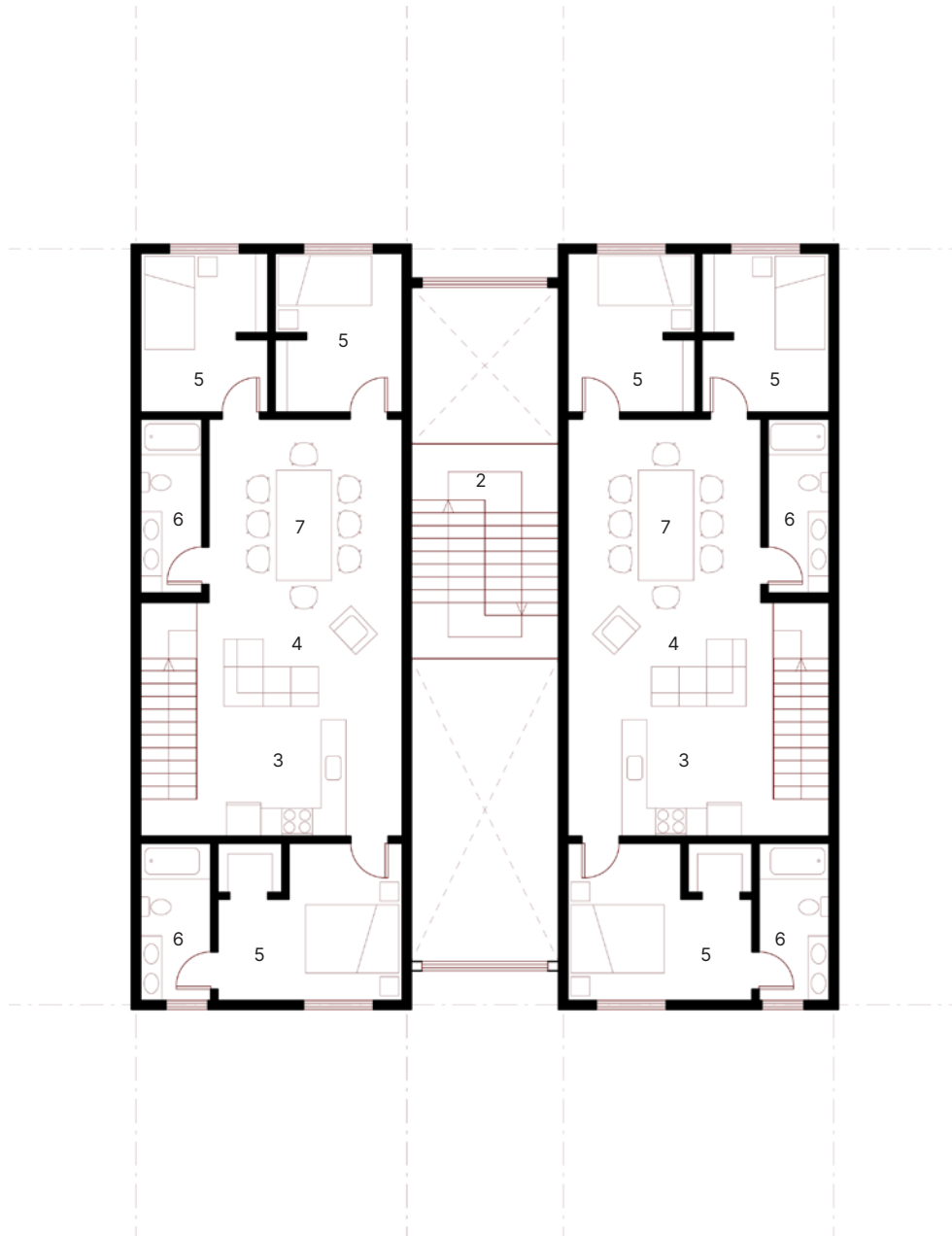
at the bottom of the townhome unit, creating an intimate walkway between the family apartments and existing row houses.

The conceptual section diagram (Fig. B) highlights the proposed unit ownership layout that will support the lease to homeownership model while also aligning with the conceptual massing strategy for the site.



- 1 Shared Entryway
- 2 Shared Stair
- 3 Kitchen/Dining
- 4 Living
- 5 Bedroom
- 6 Bathroom

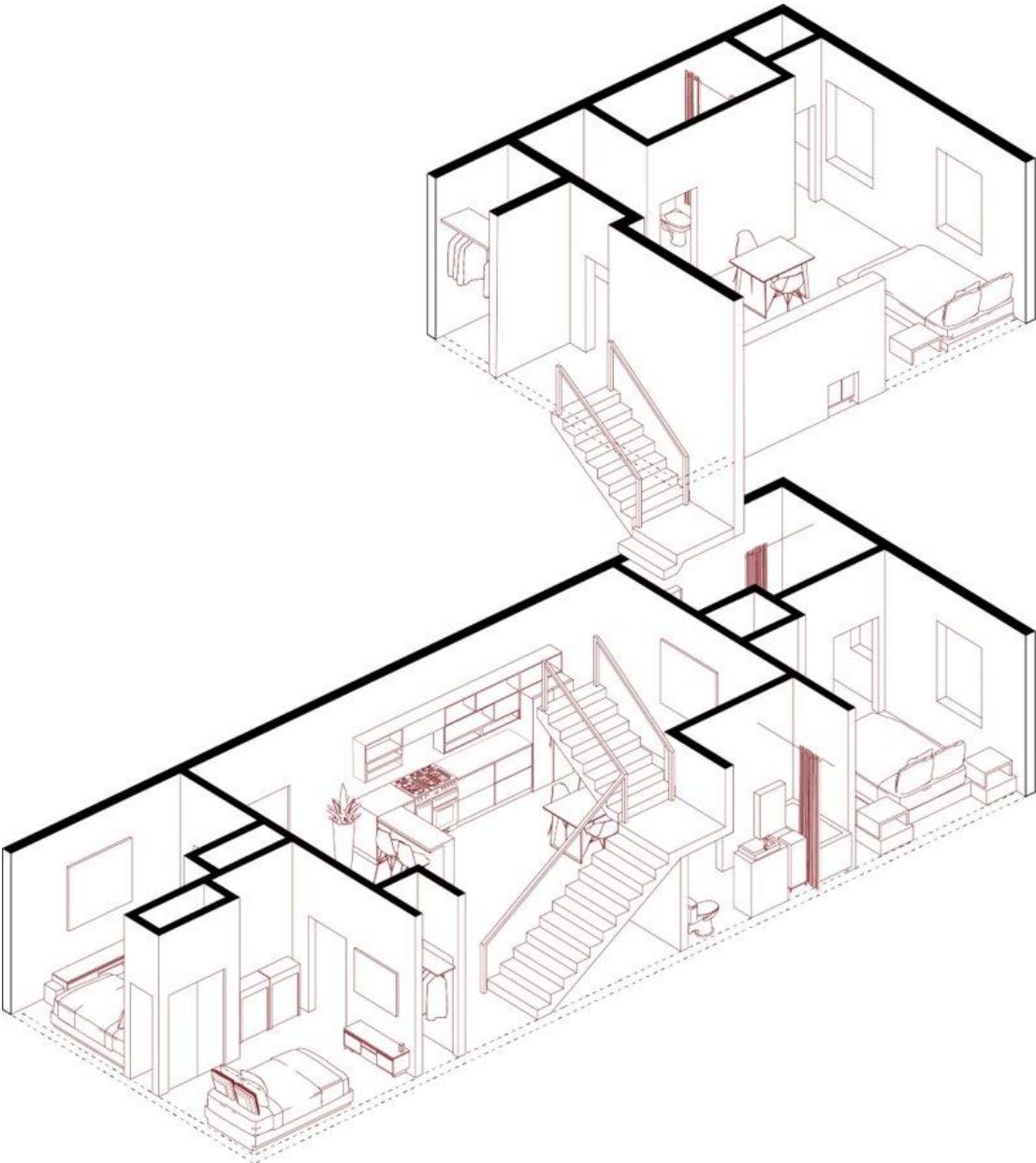




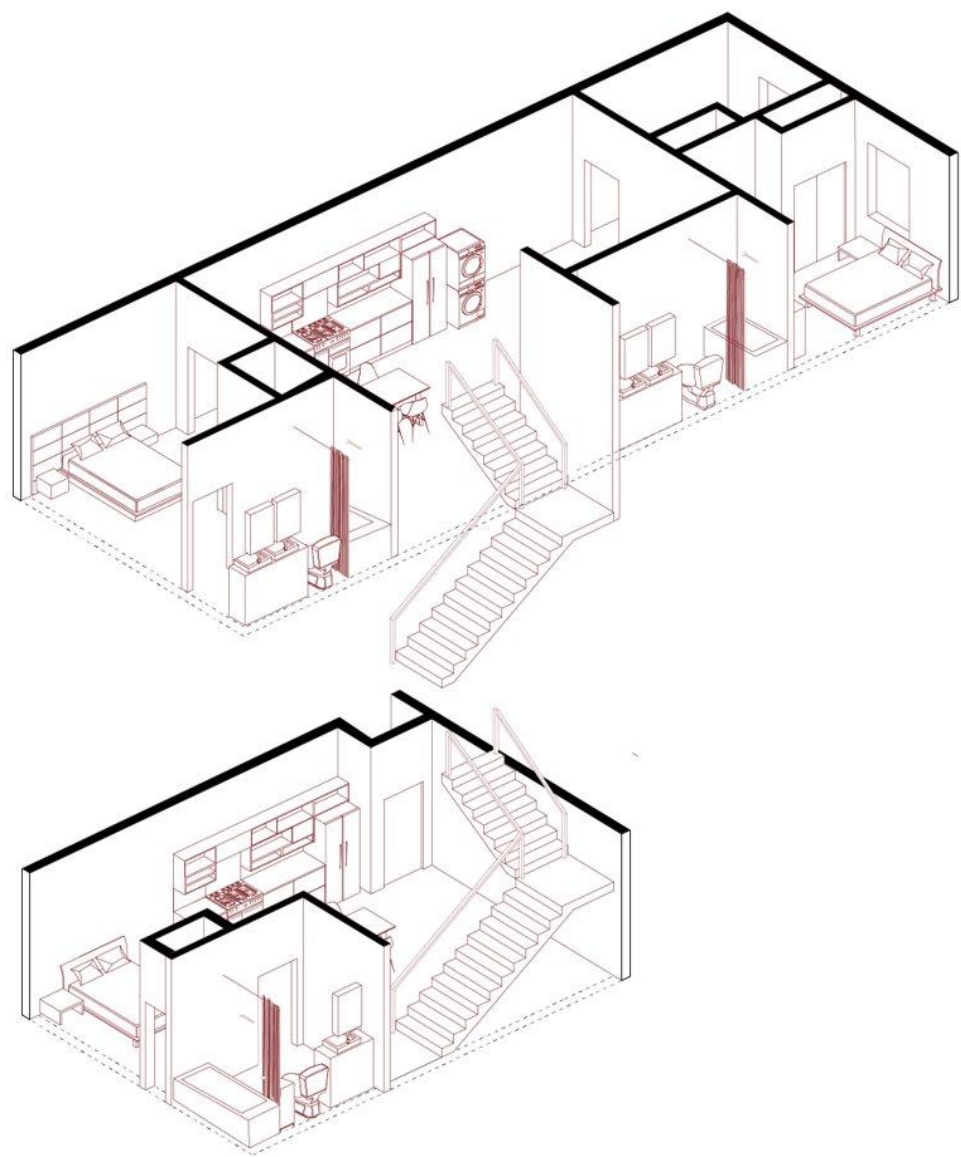
- 1 Shared Entrway
- 2 Shared Stair
- 3 Kitchen
- 4 Living
- 5 Bedroom
- 6 Bathroom
- 7 Dining



- 1 Entryway
- 2 Car Port
- 3 Living/Dining/Kitchen
- 4 Bathroom



UNIT B



UNIT MIX

Affordable Housing Units	Studio		1 Bedroom		2 Bedroom		3 Bedroom		Total
	Units	Rent	Units	Rent	Units	Rent	Units	Rent	
30% AMI (Deeply Affordable)	–	–	2	\$581	–	–	4	\$800	6
50% AMI	–	–	4	\$1004	–	–	8	\$1386	12
60% AMI	–	–	3	\$1215	–	–	6	\$1679	9
Total	0		9				18		27

Financial Strategy

The project adapts and proves the Cleveland model using 4% LIHTC credits, with the aim of scaling the model by creating 9% LIHTC set-asides similar to how Ohio does. The model targets residents at the 30%, 50%, and 60% AMI levels, which makes it 100% affordable housing.

The project draws inspiration from CHN Housing Partners, a Cleveland-based non-profit affordable housing developer. Their flagship lease purchase program develops single family homes using LIHTC, HOME funds, and traditional bank loans. Their program takes advantage of Ohio's 9% LIHTC set asides for eventual tenant ownership projects. They finance their homes with as little debt as possible, to reduce the mortgage of the tenant purchaser. They offer a loan fund to tenants who are unable to access traditional financing. The homes sell for \$20 - 25,000. They engage residents in property maintenance to prepare them for homeowner responsibilities. In exploring this model, we leaned heavily on the work and research of YLS Clinical Fellow, Adam Cowing.

Given that site control is expected through a ground lease or co-development, the assumed land acquisition cost is \$1. For hard costs, it is assumed \$200 per square foot of liveable square footage. An industry standard of 10% contingency was assumed. Financials from a similar recent development were used to estimate soft costs. They come out to 19%

of development costs which are slightly higher than expected.

With LIHTC, the tenant ownership transition in the project would begin after year fifteen. From years 0-15, Beulah Land would own the garages, garage units, lower level townhomes, and upper level townhomes. Beulah Land and the Housing Authority of New Haven would manage the shared spaces including the driveway, parking lot, stairwells, and roof. At year sixteen, Beulah Land would continue to own the garage units, offering the first right of refusal of the lower level townhome and garage to the first level tenant. The first right of refusal for the upper-level townhomes would also go to their respective units. Beulah Land would receive rental income from the garage units. In order to maintain affordability, the townhome deeds would give Beulah Land the first right of refusal to repurchase the townhomes should the tenant decide to sell.

Beulah Land would also offer some ongoing programs and classes to support those entering into home-ownership, house-hold maintenance, as well as community building amongst other topics.

SOURCES AND USES

Sources		Uses	
DOH Course Funding [a]	\$1,000,000	Land Acquisition [d]	\$1
DECD Community Investment Fund [b]	\$1,750,000	Closing Costs	\$0
4% LIHTC [c]	\$2,702,329	Hard Costs [e]	\$6,120,000
Construction Permanent Loan	\$2,035,696	Soft Costs [f]	\$1,664,082
		Contingency [g]	\$778,408
		Capitalized Interest	\$280,031
		Construction Loan Fees	\$20,357
		Operating Expenses Shortfalls	\$74,856
Total	\$7,448,025	Total	\$8,937,735

Assumptions

[a] DOH Course Funding: Assume an even split of the \$3M allocated to the Housing Connecticut course by the Department of Housing

[b] Community Investment Fund: Checked against Believe In Me Empowerment Corporation (similar development), which was awarded \$2.1M

[c] 4% LIHTC:

- i. Ineligible Items (Interest): \$280,031
- ii. Eligible Basis: 8,657,703
- iii. Applicable Fraction: 100%
- iv. Current 4% Rate: 3.43%
- v. Tax Credit Factor: \$0.91

[d] Land Acquisition Price: Assume ground lease or co-development with the city

[e] Hard Costs: \$200 per square foot based on 340

[f] Dixwell comparable development

[g] Soft Costs: 19% of development costs based on Dixwell Plaza comparable development
Contingency: Assume 10%

Revenue		2024
	<i>Rent</i>	
1BR Rent for 30% AMI	\$591	\$14,467.68
1BR Rent for 50% AMI	\$986	\$48,274.56
1BR Rent for 60% AMI	\$1183	\$43,439.76
3BR Rent for 30% AMI	\$760	\$37,209.6
3BR Rent for 50% AMI	\$1267	\$124,064.64
3BR Rent for 60% AMI	\$1521	\$111,702.24
Gross Rent Revenue		\$379,158.48
Less Vacancy	5%	\$360,200.556
Less Downpayment Set-Aside	7.5%	\$339,727.32
Expenses		
Operating expense per unit	\$7500	-208,575
Capital reserves per unit per yr	\$400	-11,124
NOI		140,501.556
Debt Service	2.205	-63,733.010
<i>Principal Loan Remaining</i>		1,979,946
Taxes	\$1000	-27,810
Cash flow after financing		\$48,958.546

Cap Rate	5.5%
Property Value	\$2,554,574
Interest Rate	5.75%
Term in years	30
Loan To Value	65%
Down Payment	\$650,000
Loan Amount	\$875,000
Annual Debt Service	\$61,876.709



*A mews-like driveway
extends through the
site .*



Neighborhood:

Fair Haven

Fair Haven began as a small oystering village to the east of New Haven's original nine squares. In the late 19th and early 20th centuries, industry came to Fair Haven and New Haven's streetcar public transit system extended into the neighborhood. This led to a denser, more mixed-use urban fabric along Grand Avenue in particular, which emerged as the neighborhood's main thoroughfare.

Fair Haven was not a main target of urban renewal, and the result is a streetscape that retains more of its density than many other New Haven neighborhoods. Residential blocks are filled with worker housing typical of the region, with brick and wood siding both being common in residential and commercial structures. Mixed-use storefronts abut the street along Grand Avenue, enhancing an active retail scene which reflects the diversity of the neighborhood.

At present Fair Haven is 64% Latino, 20% Black and 15% non-Hispanic Caucasian. This compares to city-wide totals of 30%, 31% and 30% respectively. Within Fair Haven's Hispanic population, the most common places of origin are Puerto Rico (34% of the neighborhood as a whole) and Mexico (14% of the neighborhood as a whole).

Residents benefit from an array of amenities along Grand Avenue, including two schools, phar-

macies, a grocery store, a library and many small businesses.

Grand Avenue is also the central node for transit within Fair Haven. It is the anchor of the 212 route, which currently stops at the corner of Grand & Ferry. That intersection has also been identified by the state as an eventual mini-hub for a rapid bus route. Access to transit will be vital to a denser Grand Avenue. Thanks to the neighborhood's central location between downtown and the eastern suburbs, connections like these ensure residents have access to convenient employment and retail opportunities throughout the area, and makes the corridor ripe for Transit-Oriented Developments (TODs).

Despite these strengths, the neighborhood has also suffered from the decline of industry and the displacement of the city's streetcars by car-oriented urban planning. In addition, the neighborhood has been targeted for "blight removal" in the latter half of the 20th century. These factors have contributed to a proliferation of empty space and surface parking. In 2000, the Neighborhood Plan for HUD's Empowerment Zone initiative identified the presence of suburban-style uses, like strip-mall-style retail and parking (particularly at the intersection of Grand and Ferry), as a barrier to a more walkable and cohesive experience.



A



B

Community members have shared the fact that expanses of dead space have deprived the neighborhood of safe, supportive places for young people. Concerns have also been raised about people facing crises like homelessness and addiction, drawn to these spaces because they may not be connected to supportive services.

Contributing to some of the insecurity faced by some members of the community are the overlapping challenges of poverty and inadequate housing supply. Fair Haven's poverty rate and rate of low-income households are higher than city averages, and 63% of the neighborhood's renters are rent-burdened.

Closely related to the affordability problem is a lack of adequate family housing in particular, with community members noting the health risks that come along with large families living in "the smallest of houses." Fair Haven's average household size is 3.46, compared to 2.5 for the city as a whole.

The demographic and physical makeup of Fair Haven suggests that there is significant room to build further density and connection within the neighborhood, particularly along the Grand Avenue corridor. Access to affordable housing – particularly units large enough to support families and units designed to house homeless populations and individuals suffering from substance abuse problems – are key neighborhood needs.

Housing along Grand Avenue can go beyond simply fulfilling a

need, however, and play a key role in further stitching together the neighborhood's principal commercial corridor and central node for civic gathering.

Fig. A: Fairhaven today is full of restaurants and bakeries that cater to the neighborhood's residents. Source: Arnold Gold / Hearst Connecticut Media

Fig. B: Fairhaven in 1927 at the intersection of Grand Ave and Franklin.

PROJECT:

GRAND AVE

The project proposes a new 150-unit, mixed-use development in the heart of Fair Haven that will address demand for additional affordable housing while contributing public space and community uses to the enhancing the walkability and dynamism of the neighborhood's main artery, Grand Avenue. The ambition of this This first, strategic development is that it might can catalyze additional in-fill projects that are more responsive to the corridor's potential for denser, mixed-use development. In the process, the project can expand opportunity and shape a built environment that better serves the residents of Fair Haven.

Team members

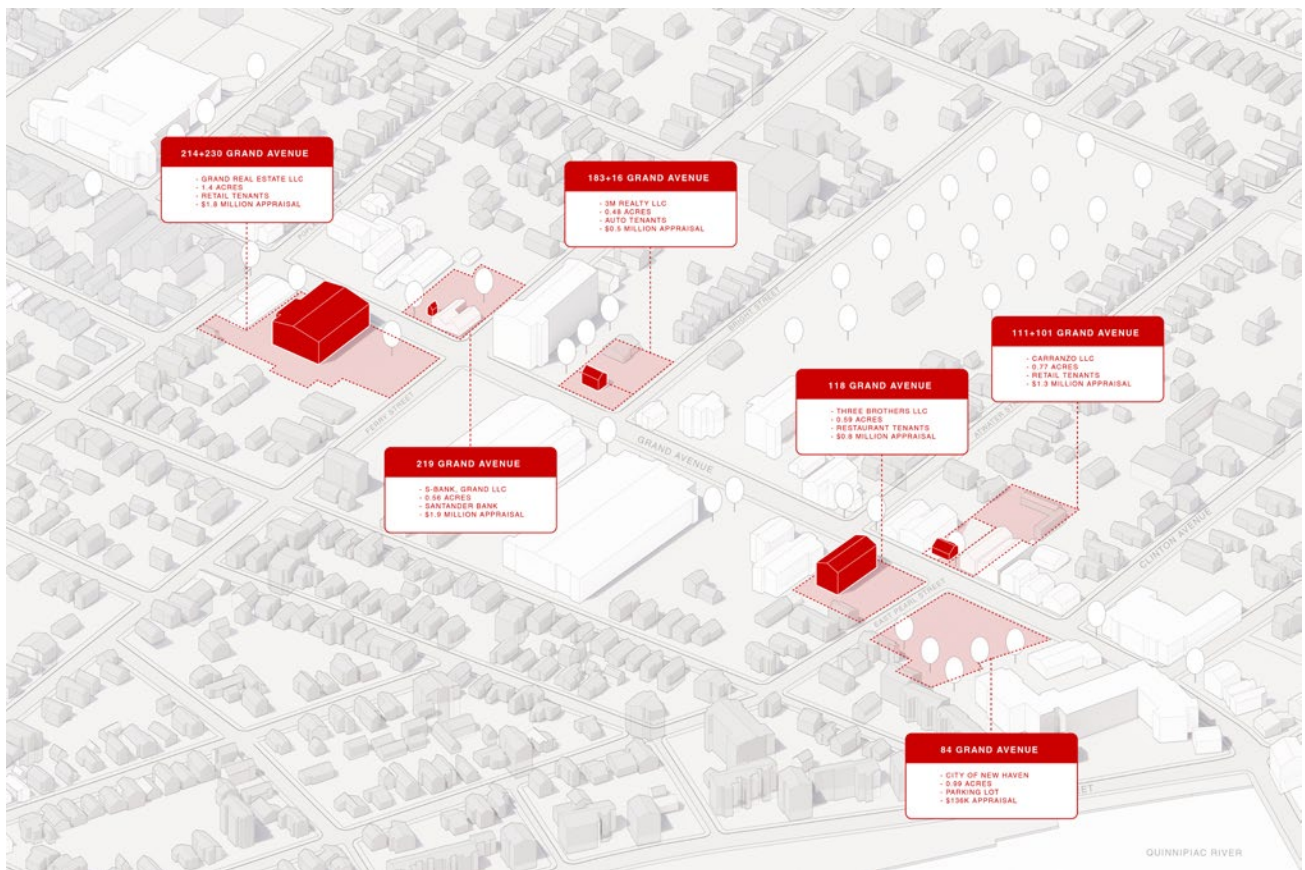
School of Management: Malcolm Davis, Dahlia Leffell

School of Law: Nicholas Fernandez, Gabriel Gassmann

School of Architecture: Serena Liu, Eric Wang, Annika Babra, Jiaying Yan

in collaboration with

Neighborworks New Horizons



A

Context

Client

NeighborWorks New Horizons (NWNH) is a Community Housing Development Organization based in Fair Haven. NWNH has served south central Connecticut for thirty years, developing and managing over 650 residential units and 12 commercial units. NWNH also offers services to residents including homeownership counseling, financial literacy classes, youth programs and resident engagement programs.

NWNH is most experienced in single-use multifamily housing but has several mixed-use properties, including 255-259 Grand Avenue (just yards away from the Grand & Ferry site). They manage approximately 100 units in Fair Haven and have demonstrated an interest in block-wide development as a strategy for neighborhood improvement.

Methodology

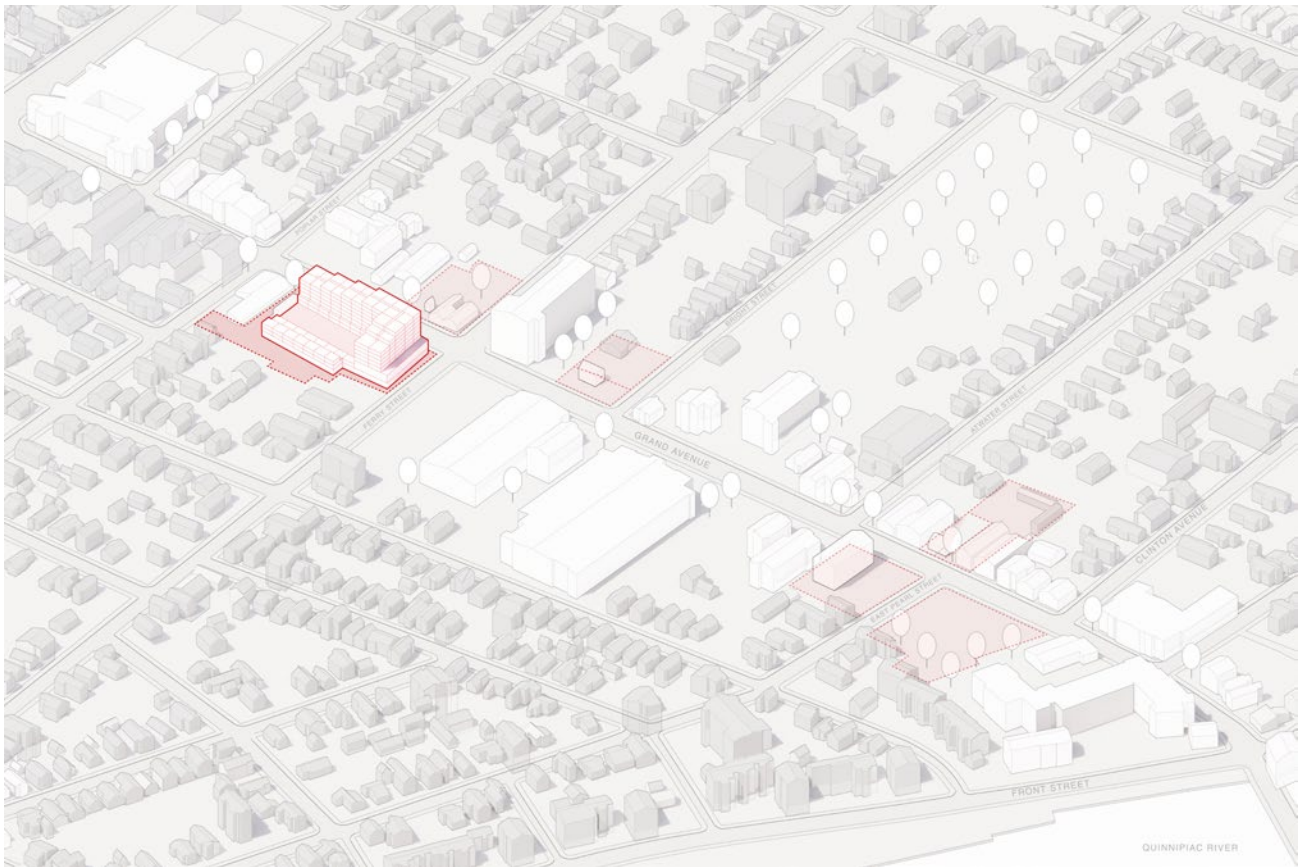
Members of the team spoke with the following community members: Suzanne Lagarde, CEO of Fair Haven Community Health Care (September 29); Dave Greco, co-founder of ARTE Inc. (September 30); and Diane Elton and Lee Cruz, members of Fair Haven Community Management Team (October 3).

The team sought information from the following parties: Con-

necticut Green Bank; Giovanni Zinn, City Engineer; and Tony Schaffer, C.A. White, Inc. We also consulted the New Haven Free Public Library's local history collection to review the 2000 Neighborhood Plan (required by HUD for the city's Empowerment Zone application) and the 1969 Redevelopment and Renewal Plan (produced for the city's federal urban renewal program).

Demographic data for our final presentation was sourced from DataHaven, a nonprofit providing Connecticut-specific neighborhood information. Maps for cost-burdened households and average household size were sourced from Policy Map, a national data "warehouse" and "analytics platform."

Fig. A Grand Ave in Fairhaven provided many possible opportunities for development.



B

Proposal

Neighborhood Strategy

At its core, this development aspires to address the lack of affordable housing in the area, helping mitigate the burden of living costs by providing a typology that mixes affordable townhouses, supportive housing through SRO units, and one- to two-bedroom market-rate units. The development aspires to would also reinforce the site's current status as a critical junction for transit, spontaneous gathering and civic expression through features like a public plaza and its contribution to a more active streetscape. In doing so, the project proposal intends to both demonstrate the viability of mixed-use, mixed-income development in Fair Haven and provide a template for carrying out similar projects in the neighborhood and beyond.

Along with market-rate units commercial space provides a source of income for the project, and are central to its economic viability is also vital to the long-term economic sustainability of this project. The ground-floor retail spaces are designed to be able to host larger commercial tenants as well as to be suitable for local businesses in the area, to foster a varied and locally-oriented smaller than big-box spaces to cater to local businesses, offering opportunities for existing commercial tenants to return as well as further fostering a

dynamic and locally-oriented Grand Avenue corridor. Various community organizations can also benefit from access to the elevated courtyard and amenity spaces, as well as a close proximity to groups of people they serve.

Architectural Strategy

The proposed project is a ten-story building providing 150 units of housing and 11,900 square feet of commercial space, as well as generous green space for the residents. The idea is that the building is an ecosystem of its own, serving the diverse needs of the neighborhood. It weaves together a range of housing types, including large family units, 1-2 bedroom apartments and single-occupancy supportive housing units, as well as generous community spaces. The building is composed of a concrete podium, a row of townhomes, and a mass timber residential tower. The development seeks to address a multi-faceted set of goals, acknowledging that the site's potential lies in its ability to accommodate density strategically and at a meaningful scale.

The ground floor accommodates provides storefronts that support local businesses and offer space for community organizations. Frontage along both Ferry Street and Grand Avenue improves the pedestrian experience of both ar-

Fig. B The chosen site sits on a key corner on Grand Ave, acting as an anchor building.

SITE PLAN



teries at this intersection. A garage with abundant parking is concealed behind the retail wrapper. Providing 81 parking spots, it exceeds the requirement of Inclusionary Development zoning and alleviates traffic pressures of this dense resident community. Entry to the parking lot is located on Poplar Street to avoid additional traffic congestion on Ferry Street.

The second floor houses 15 three-bedroom townhomes and 9 one-bedroom apartments. The deck above the parking lot is populated with ample vegetation. Framed by rows of townhomes, the courtyard is an ideal setting for community gatherings and children's playtime. A residential amenity space also has direct access to the courtyard. Above the townhomes is a floor with 27 SRO units. The spaces at the corner and the ends of the building will be used as shared space for the residents of the SRO units, as well as offices for nurses, case workers, or other staff. The inclusion of supportive housing in the development tackles issues such as homelessness and addiction in the neighborhood.

One-bedroom and two-bedroom apartments occupy the fifth to tenth floors. There are 108 apartment units organized along double loaded corridors, all served by one elevator core. The building steps in 2 units every floor from the eighth floor. The two vertical fire stairs are connected by an additional staircase running along the top floors of the building. The terraces can be green roofs or solar energy farms, increasing the overall

energy efficiency and stormwater capture capacity. Adding the aforementioned elevated courtyard, the development adds a great amount of greenery to the neighborhood.

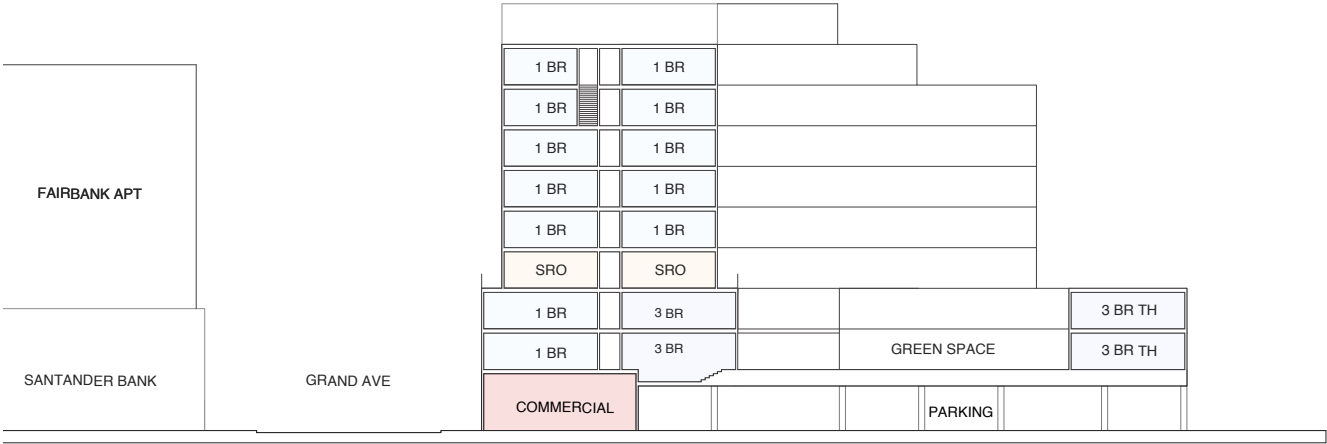
The mixed-typology design seeks to encourage a sense of community within the building itself. The development is sited around a private greenspace for residents, located on the second floor and faced by family townhouses. This area helps address health and safety concerns thanks to its greenery and relative containment. It can also act as a flexible space for youth, after-school activities and even public community events if accessed through adjacent amenity space. Additionally, the floor of SRO units addresses the need for supportive housing. It is designed so that this critical housing is not segregated from the wider community, but has access to essential services, private outdoor space, and care workers within the building.

By foregrounding sustainable design through mass timber, energy efficiency through a stepped typology allowing south-facing solar panels, and potential gray and stormwater management, this proposal acts in alignment with New Haven Vision 2025 and the state's greater goals of providing quality, healthy, and energy-efficient spaces.



COMMERCIAL/
PARKING

SECTION

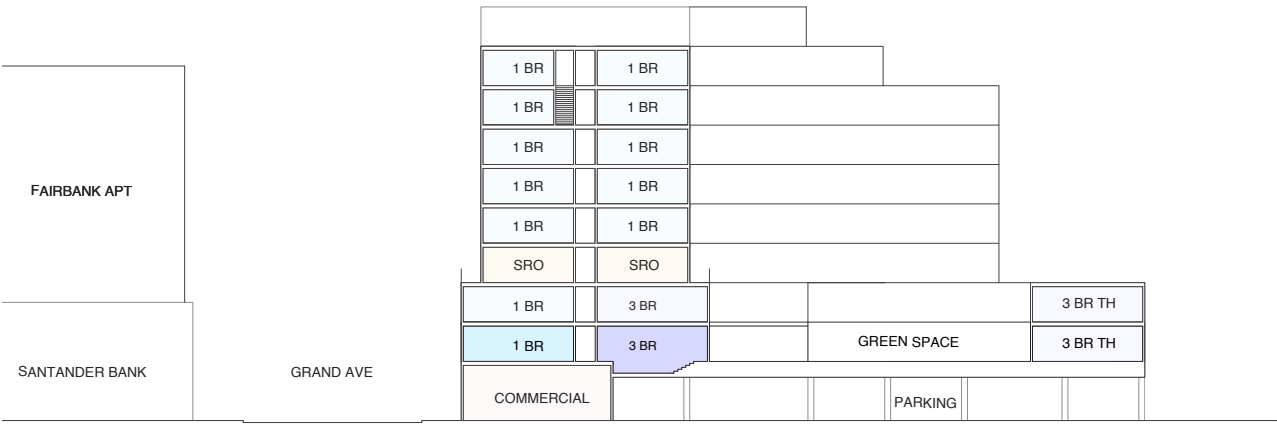




At the building's corner, a public plaza enhances and celebrates the spontaneous interaction and gathering at this important intersection. The plaza aims to create a new civic space for the neighborhood. Accentuating the space with murals, perhaps from Fair Haven's Latino community, would be a meaningful way to deepen a sense of place and connection. The main residential entry faces this plaza, contributing to the liveliness of this community.



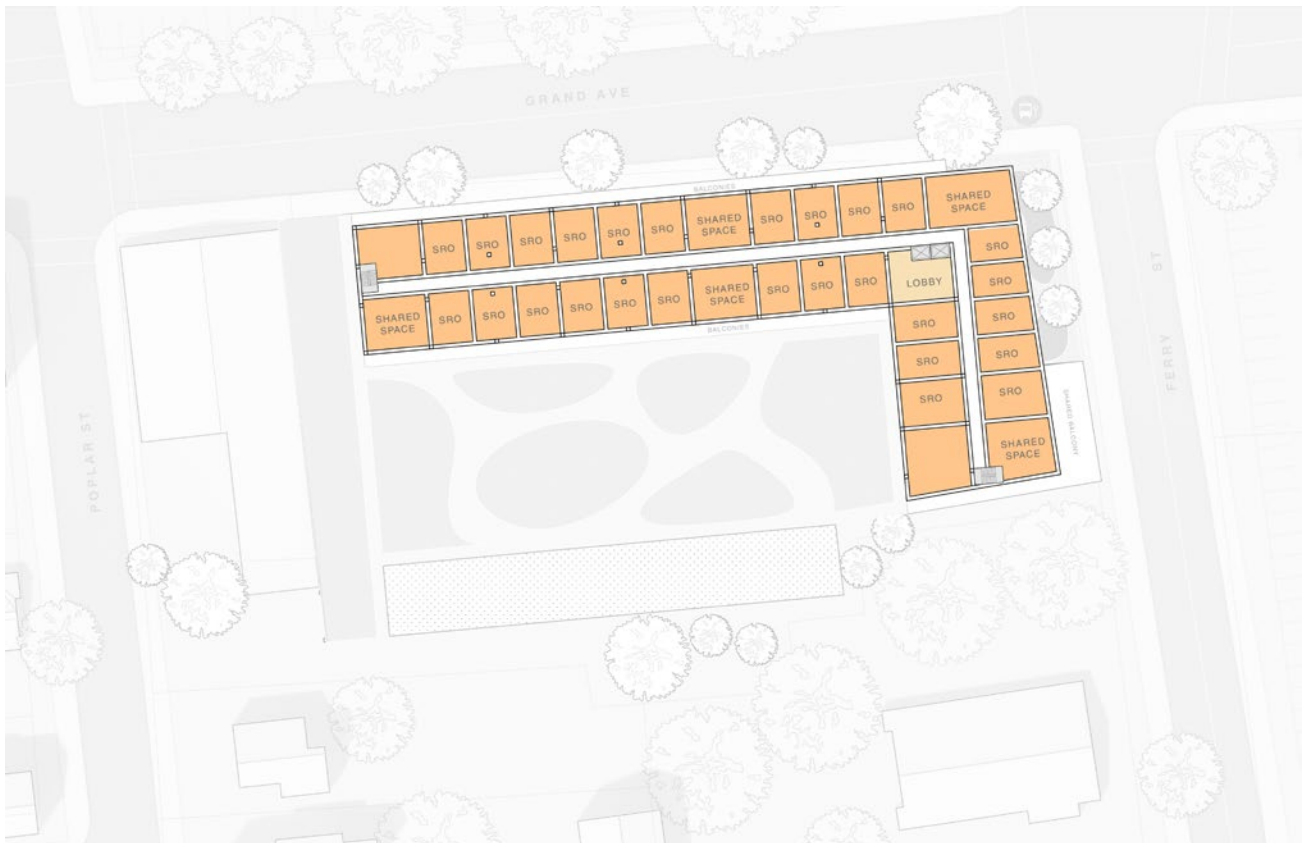


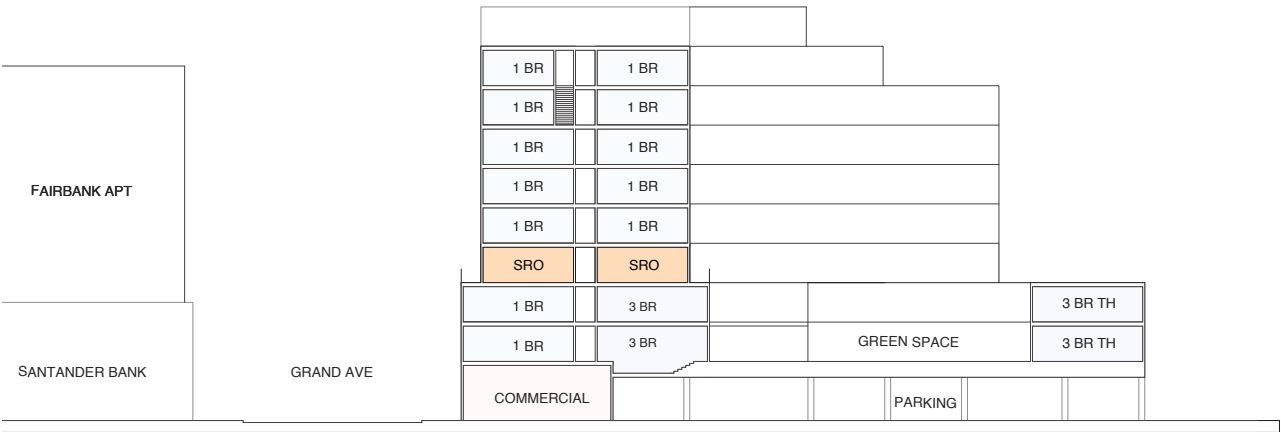




The project's design accounts for solar angles, ensuring the courtyard receives light throughout the year.



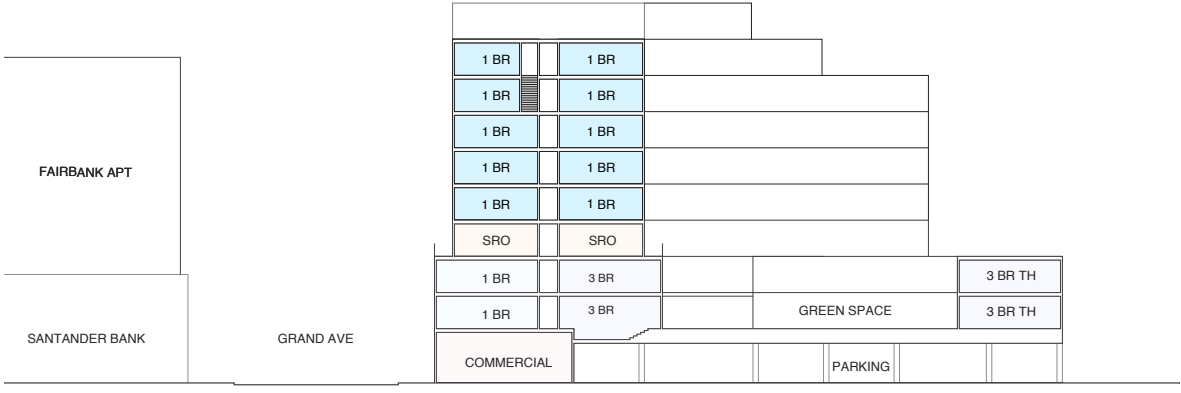






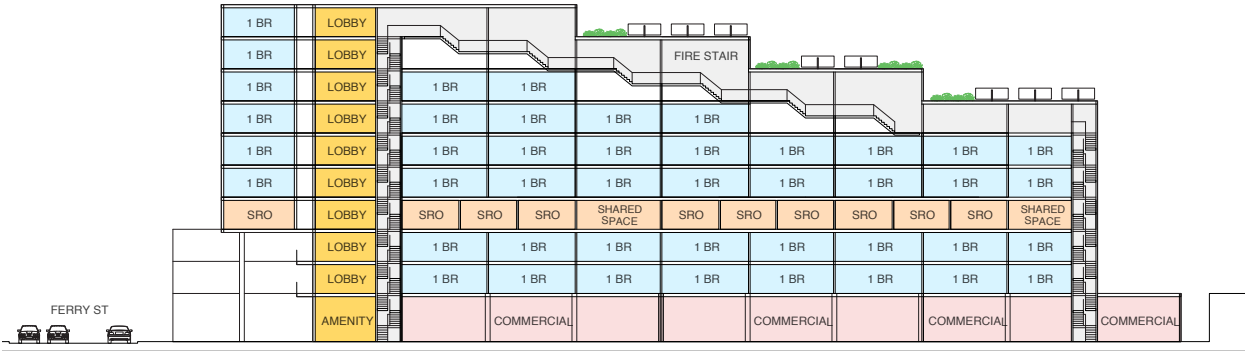
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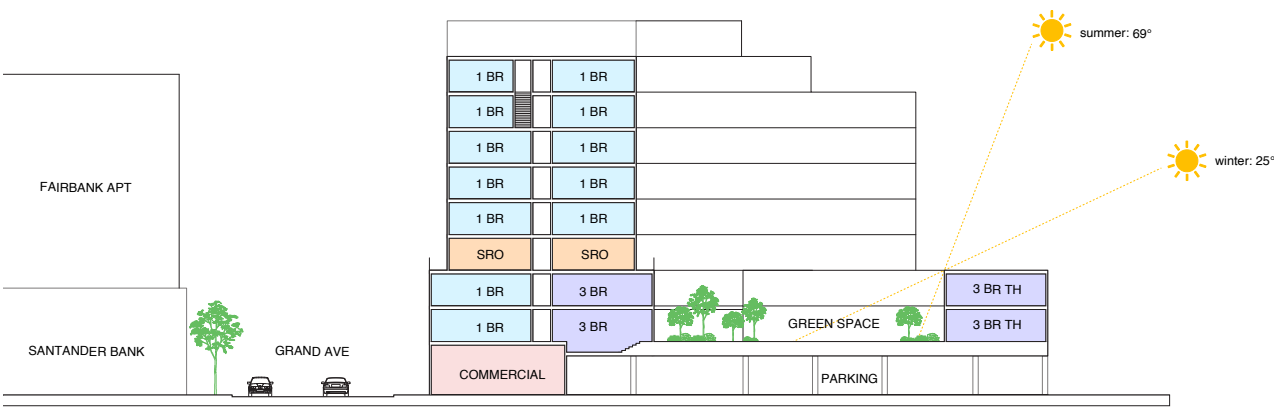
SECTION



SOUTH

SECTION



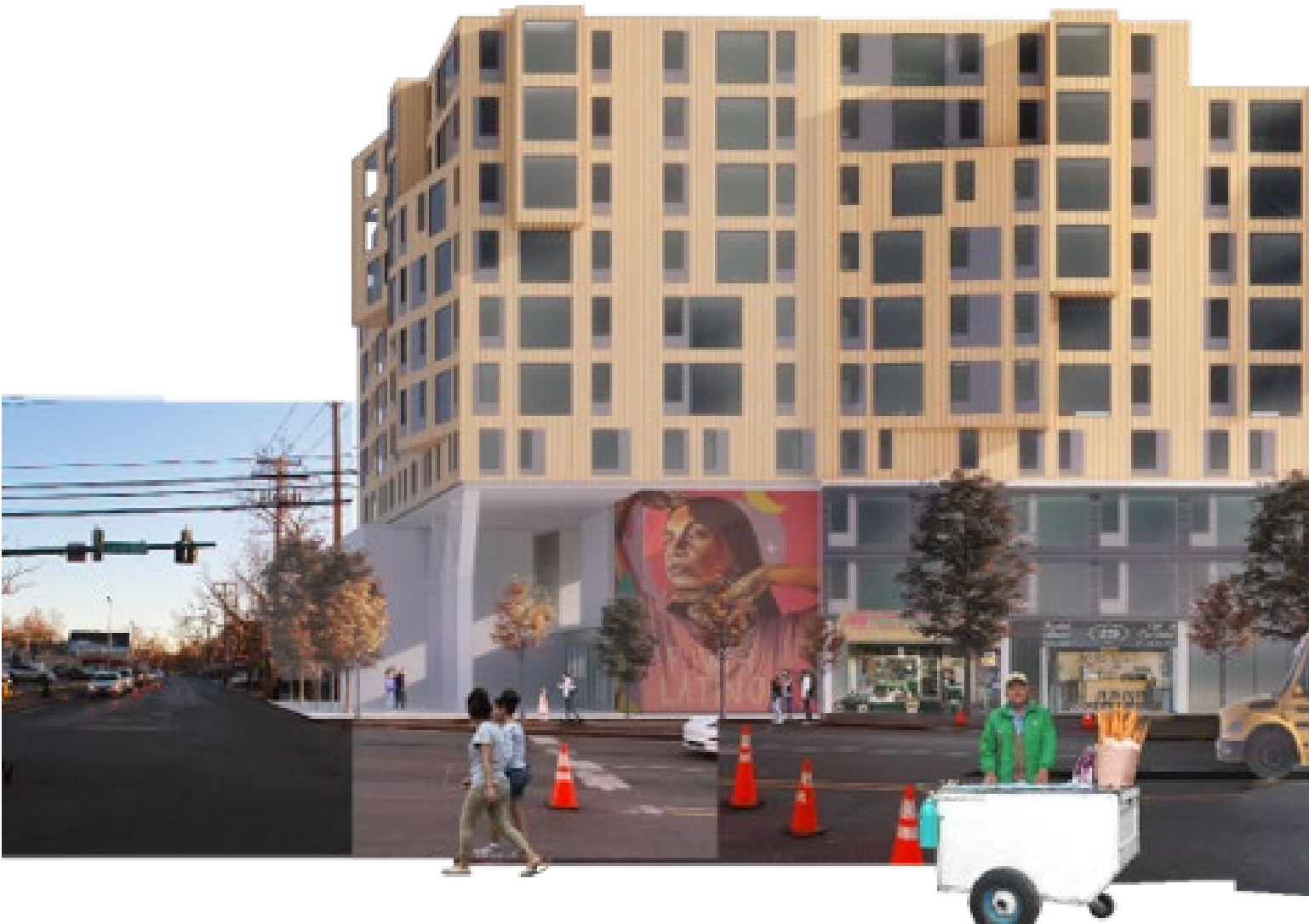




Commercial street frontage provides a source of income for the project. The ground-floor retail spaces are designed to be able to host larger commercial tenants as well as to be suitable for local businesses in the area.

The height of the building mimics the Fairbank Apartments, across the intersection of Grand Avenue and Ferry Street. The diagonal locations of the towers ensure there will not be a canyon effect where sunlight and view are severely blocked by groups of tall buildings.







UNIT MIX

Affordable Housing Units	Studio		1 Bedroom		2 Bedroom		3 Bedroom		Total
	Units	Rent	Units	Rent	Units	Rent	Units	Rent	
30% AMI (Deeply Affordable)	27	\$591	–	–	–	–	5	\$878	32
50% AMI	–	–	–	–	48	\$1267	10	\$1464	58
60% AMI	–	–	40	\$1690	20	\$2028	–	–	60
Total	27		9		68		15		150

Financial Strategy

The financial model for this development is preliminary, and for a project of this size, much is subject to change. There are five primary areas that could have the largest impact on the model should they change. The areas are: construction costs, market rate demand, interest rates, commercial rents and vacancy rates, and phasing of construction.

Construction costs: Due to the pandemic and rising inflation, construction costs have been unpredictable for the last several years. Additionally, NWNH had not completed a comparable tower project that could have been used to extrapolate costs. Our current construction cost estimates (\$330 per square foot) would need to be further interrogated given the lack of high-rise buildings being constructed in New Haven.

Furthermore, the proposal plans for a mass timber/CLT construction system in order to provide more flexibility, sustainability, and warmth to our design. However, as this is a newer technology, the costs associated with the material, and the oversight it requires, could increase significantly from current estimates. A way to mitigate this risk would be using more traditional steel and concrete, though potentially less sustainable.

Market rate demand: The current estimate assumes that the demand for market-rate units will only produce a 5.5% vacancy rate. This assumption is based on the Department of Housing Consoli-

dated Application workbook. The project will introduce 60 new 1- and 2-bedroom market rate units to the Fair Haven market. This is a significant addition, and a more in-depth market study is required to confirm that the vacancy assumption is reasonable. Should demand prove unsupportive of this number of units, NWNH might consider making more of them affordable at the 50-60% AMI level, as there is certainly demand for affordable housing in the neighborhood.

Interest rates: Given the current economic environment, interest rates are at a very high level. As the project progresses, uncertainty in the economy could lead to even higher interest rates, which would affect the project's ability to carry debt. The developer would want to be strategic in how much debt is taken on (and when), in order to optimize the amount of interest they will need to pay. Taking a phased approach to development (as discussed below) might be one strategy for defraying the cost of capital and delaying until interest rates come down. They may also want to consider taking on less debt than the project could potentially carry, in order to minimize these costs, though this will of course depend on the other funding available to them.

Commercial rents and vacancy rates: The current design includes 11,900 square feet of commercial space. The model assumes that one third of that space will be rented by an anchor tenant at a rate of \$10 per square foot. The remaining two thirds will

SOURCES AND USES

Sources		Uses	
DOH Course Funding [a]	\$1,000,000	Land Acquisition [d]	\$1,789,100
DOH Bonds	\$4,000,000	Closing Costs [e]	\$17,891
DECD Community Investment Fund	\$29,518,356	Hard Costs [f]	\$57,765,000
4% LIHTC [b]	\$14,341,478	Soft Costs [g]	\$10,053,198
HTCC [c]	\$1,500,000	Contingency [h]	\$6,781,820
Construction/Permanent Loan	\$30,077,938	Capitalized Interest	\$3,081,378
		Construction Loan Fees	\$300,779
		Operating Expenses Shortfalls	\$648,606
Total	\$80,437,772	Total	\$80,437,772

Assumptions

[a] DOH Course Funding: Assume an even split of the \$3M allocated to the Housing Connecticut course by the Department of Housing

[b] 4% LIHTC:

- i. Ineligible Items (Land, Lease Buyout, Commercial Space): \$5,981,100
- ii. Eligible Basis: \$68,252,130
- iii. High Cost Adjustment: 130%
- iv. Applicable Fraction: 48%
- v. Current 4% Rate: 3.43%
- vi. Tax Credit Factor: \$0.98

[c] HTCC: Assume awarded the limit of \$500,000 per year for 3 years

[d] Land Acquisition Price: Based on 2021 appraised value

[e] Closing Costs: 1% of land price

[f] Hard Costs:

- i. \$330 per square foot construction cost based on LaRosa Building Group estimation
- ii. \$16 per square foot to buyout leases of current retail tenants based on available documentation
- iii. \$20 per square foot for demolition costs based on contractor estimates

[g] Soft Costs: Assume 14% of Hard Costs

[h] Contingency: Assume 10%

be rented out at \$18 per square foot. These assumptions are based on estimates received from Tony Schaffer of C.A. White. The model also assumes a 10% commercial vacancy rate, which was recommended to us by Brian Murray of Shift Capital. We are being fairly conservative in our assumptions, but given that retail can be a volatile market, significant change is a risk. Working to secure an anchor tenant first, would be important to build some certainty into the model. This tenant could be a community organization or local business (potentially Long Wharf Theater, Arte Inc., or Fair Haven Community Health) that would also serve the mission of the project.

Phased construction: The financial model proposes a single phase of development and construction for the whole building. This would be advantageous from a design perspective, as the project aims to establish a cohesive community. However, given the size and cost of the building, it might be more realistic to phase construction in order to defer costs. If the project were split into two or three phases, we could apply for LIHTC, CIF, and other grant funding multiple times, therefore maximizing the amount of subsidy we receive. However, splitting the project would require significant revisions to the design of the building, so more work must be done to balance the need to maximize subsidy with the overall mission of our design.

A note on unit mix: The current unit and affordability mix was created to balance the goals of

providing family housing and supportive housing, maintaining storefront retail to enhance the walkability of Grand Avenue, qualifying for LIHTC, and having the project pencil out. During the development process, different mixes were tested, such as including more 3-bedroom units for families and pushing the affordability mix to 80-20 (therefore foregoing LIHTC). The final mix that was settled on does work out, but more work could be done to advance the goals stated above as the project is refined. For example, as costs come into clearer focus, perhaps the building will be able to support more affordable units or more larger units.

STABILIZED YEAR

CASH FLOW

Revenue		2027
<i>Rental Revenue</i>	<i>Rent</i>	
Studio Rent for 30% AMI	\$591	\$207,268.440
1BR Rent for Market Rate	\$1690	\$878,068.968
2BR Rent for 50% AMI	\$1267	\$789,950.331
2BR Rent for Market Rate	\$2028	\$526,841.381
3BR Rent for 30% AMI	\$878	\$57,022.526
3BR Rent for 50% AMI	\$1464	\$190,161.682
Gross Rent Revenue		\$2,649,313.328
Less Vacancy	6%	\$2,503,601.095
<i>Commercial Revenue</i>	<i>Rent (per sq. ft.)</i>	
Anchor Retailer	\$10	\$515281.005
Individual Retailer	\$18	\$1854777.814
Gross Commercial Revenue		\$2370058.820
Less Vacancy	10%	\$2133052.938
Total Revenue		\$4636654.033
Expenses		
<i>Residential Rental Expenses</i>		
Operating expense per unit	\$7500	\$1,266,197.411
Capital reserves per unit per yr	\$400	\$67,530.529
Residential Expenses		\$1,333,727.940
<i>Commercial Rental Expenses</i>		
Operating expense per unit	40%	\$948,023.528
Capital reserves per unit per yr		
Commercial Expenses		\$948,023.528
Total Expenses		\$2,281,751.468
NOI		\$2,354,902.565
Debt Service	1.569	\$1,501,140
Principal Loan Remaining		
Taxes	\$1000	\$150,000
Cash flow after financing		\$703,762.565

Rent Escalation	2%
Expense Escalation	3%

Cap Rate	5.5%
Property Value	\$42,816,410.269
Interest Rate	5.75%
Term in years	30
Loan To Value	65%
Down Payment	0
Loan Amount	\$35,320,867
Annual Debt Service	\$1,501,140

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